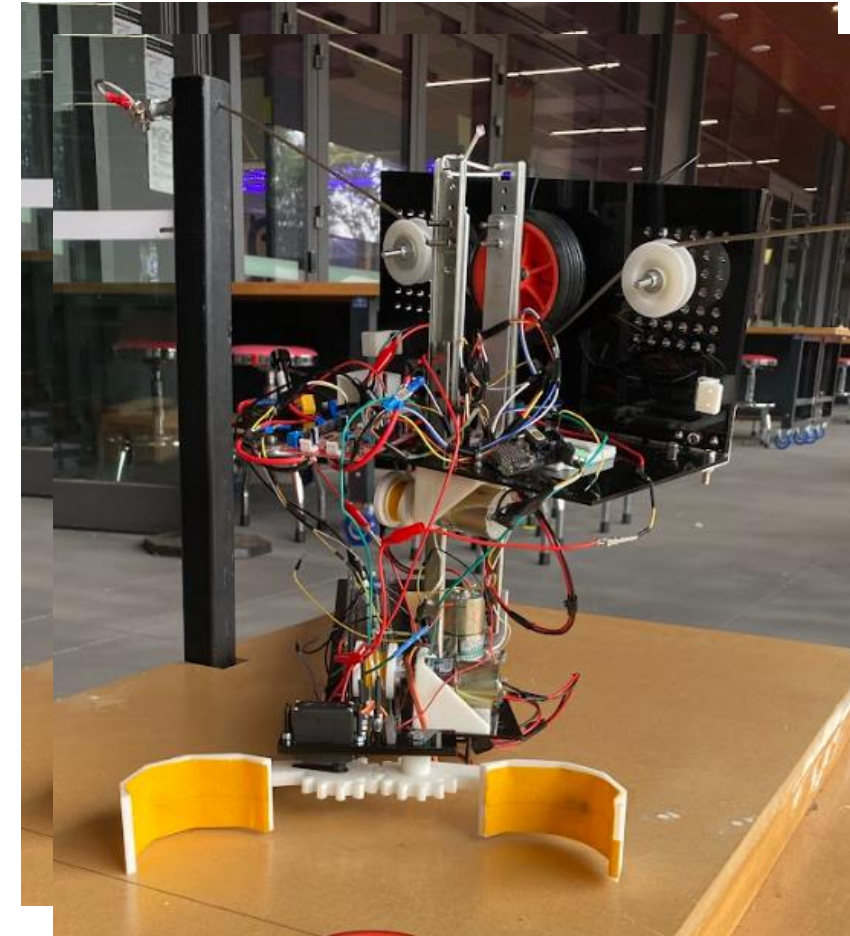
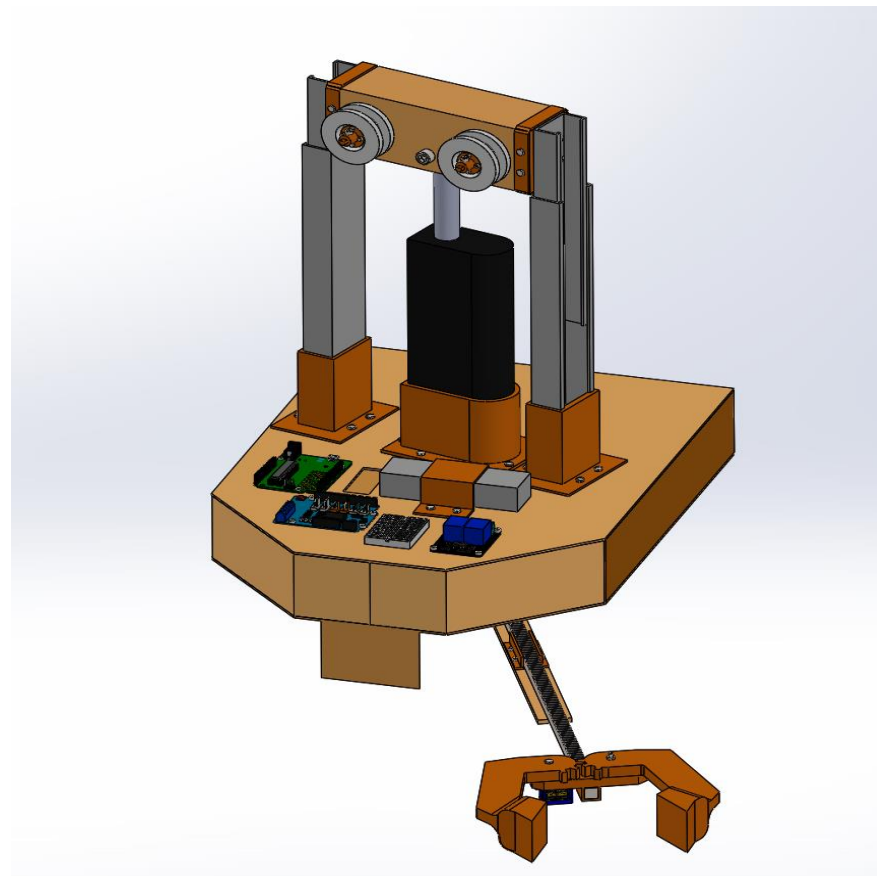


TEAM 21 - Warman Project: Final Design Submission 2022



Jason Abi Chebl
31444059



Faisal Ibrahim M Almaslukh
31042643



Ryan Hartshorne
30492165



Mahdir Shafin Hasan
32236093

i Executive Summary

The team was able to design and build a prototype of the transporter system for the client. The project was diligently completed on time, however, the team wasn't able to ensure the whole system would work as efficiently as possible - with some issues unable to be resolved promptly before delivering the final project to the client.

This report examines and analyses the conceptualisation of the system, highlighting the reasons why some aspects did not work as expected. In particular, it provides a visual and written comparison between the teams preliminary design and the final manufactured system - with the two major design changes being the use of a drawer slide mechanism instead of a linear actuator (to raise the system) and a rack and pinion (to extend the arm). In addition, a qualitative investigation was conducted and it was found that inadequate testing was the most prominent issue that caused the system to not perform as expected. The team was able to illustrate their final design using detailed drawings, reconcile all spending of the project as well as provide a recount of all team meetings conducted.

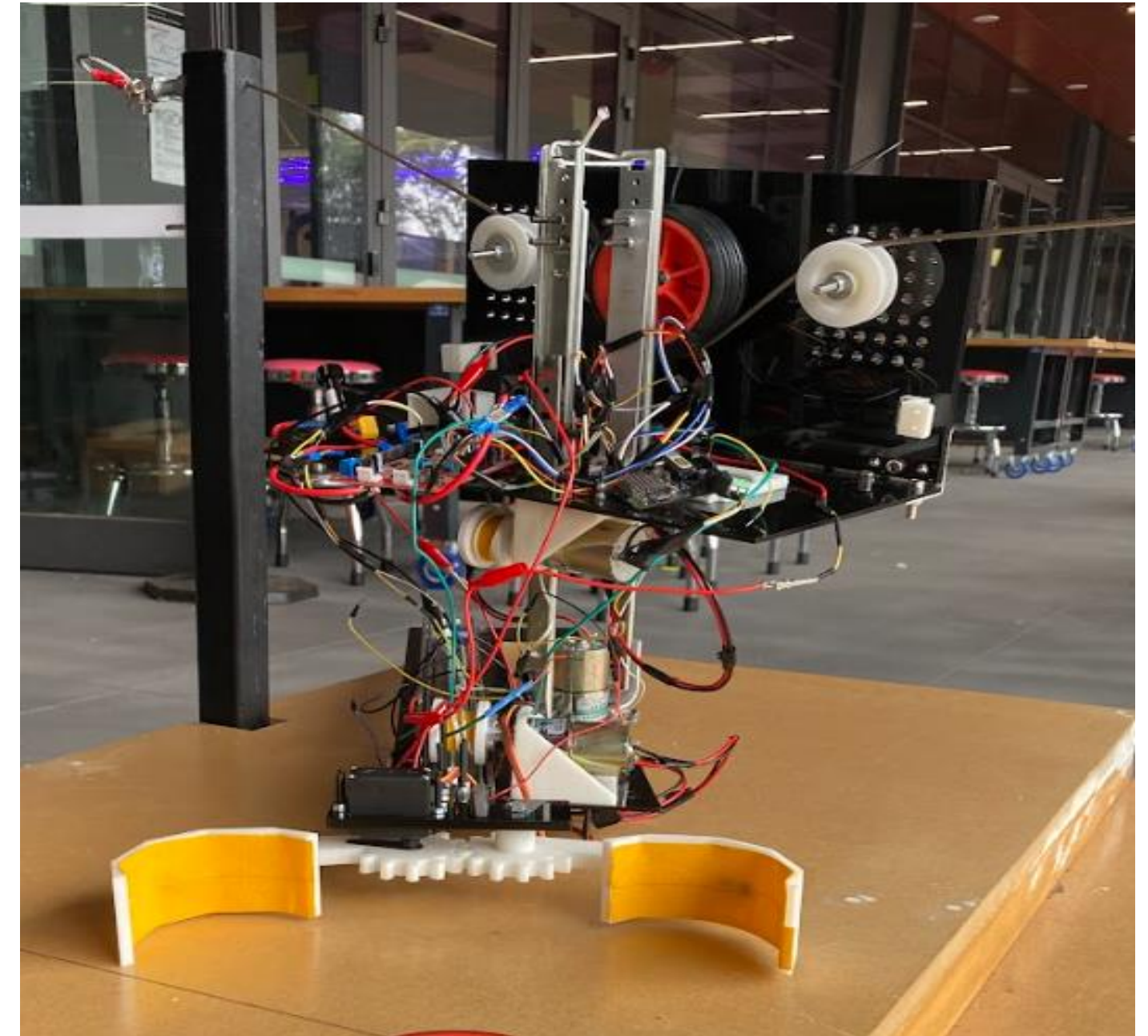


Figure 1: The final manufactured system.

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1 Introduction

The Warman Competition is a yearly competition that aims to encourage engineers to design and build a prototype that will solve a Engineering problem provided. This year, the competition's objective was to "build a scaled prototype system capable of transporting a package over a chasm zone to the designated delivery region in the fastest time possible, whilst achieving the team's personal learning goals" [1]. To tackle this challenge, our team of four engineering students, pursuing either Mechanical or Mechatronics engineering, collaborated together to build a solution. Having designed, prototyped and tested, the team competed on 10th of May and was able to secure the 33rd place in the competition.

While the team worked hard to achieve their goals, unfortunately not everything worked out as planned. Hence, the purpose of this report is to reflect on the system built and the process that the team underwent. In particular, this report outlines and highlight the two major changes that the team made from the preliminary design to the final system as well as incorporates a project reflection which focuses on the team's overall performance. The report also includes a break down of the project's expenses, four engineering detailed drawings of some of the more complex parts manufactured by the team and a record of team meetings.



Figure 2: Warman Competition Track.



Figure 3: Final Preliminary Design.

2.1 CAD Images (Set-up and ready for start)

PHASE 1

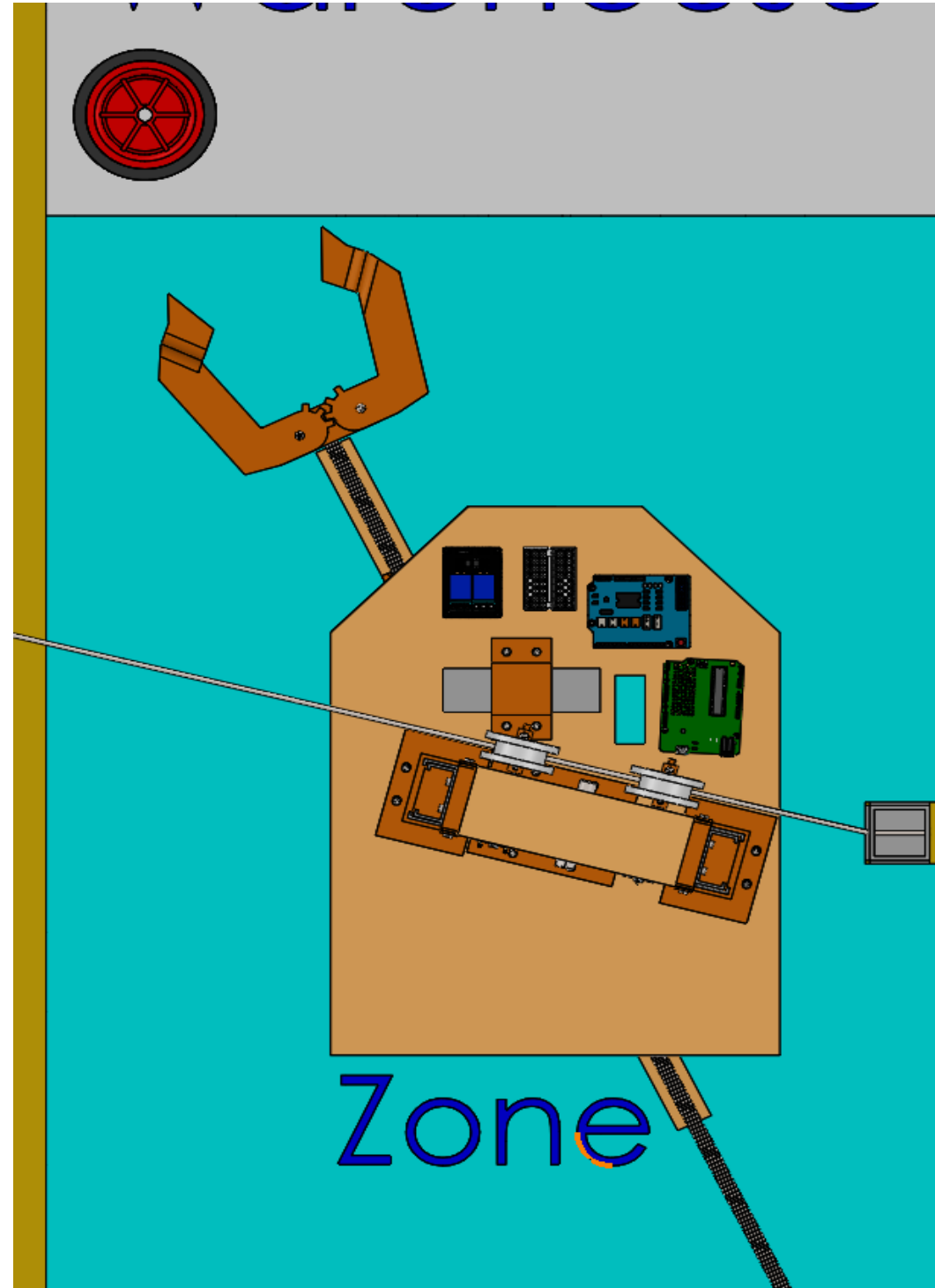


Figure 4: Top view of system in the Start Zone. The system should fit in the Start Zone.

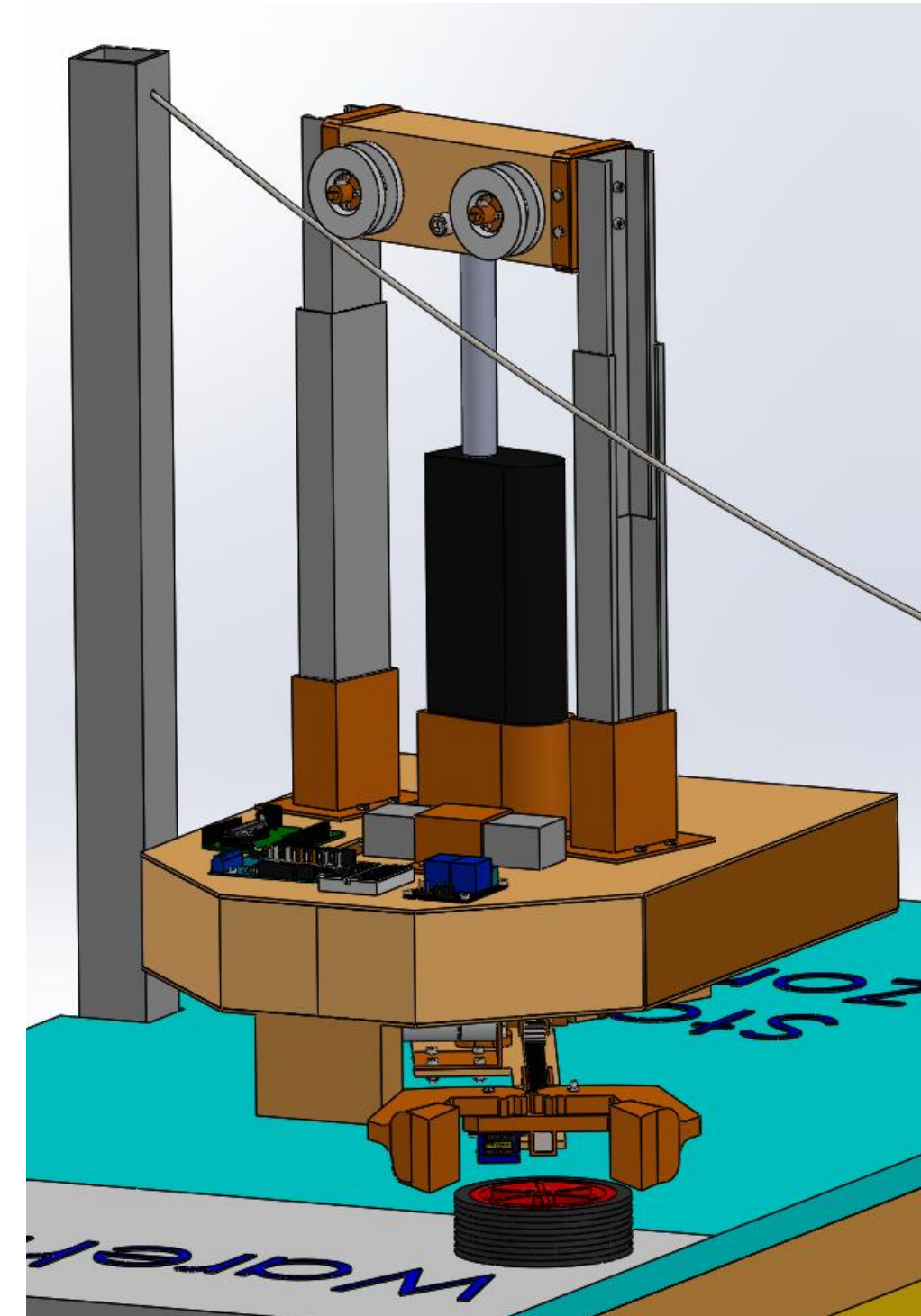


Figure 5: The system claws should be set up and aligned with the package.

2.1 CAD Images (Collecting Payload)

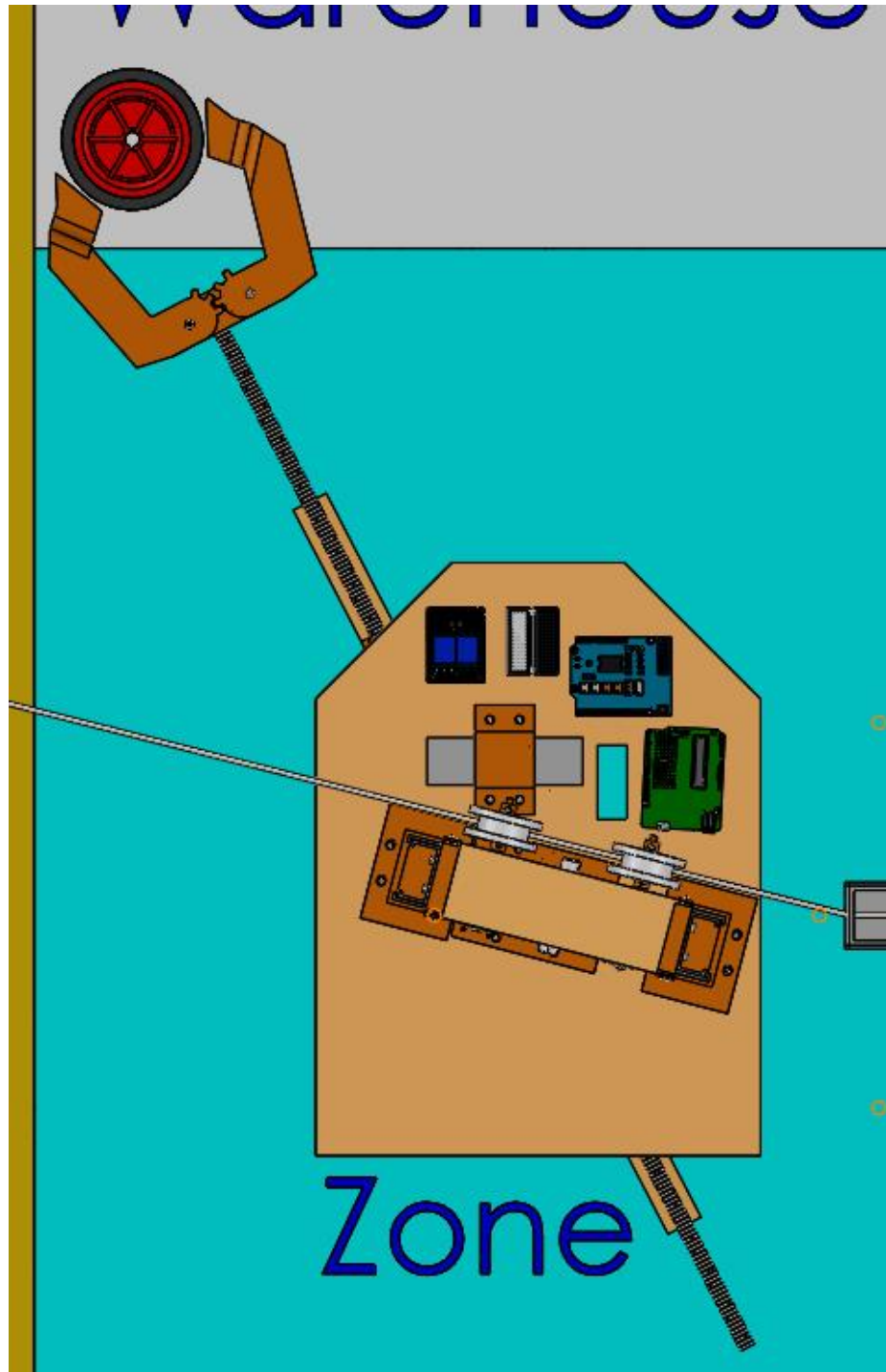


Figure 6: The system will have a horizontal arm that reaches out to the package.

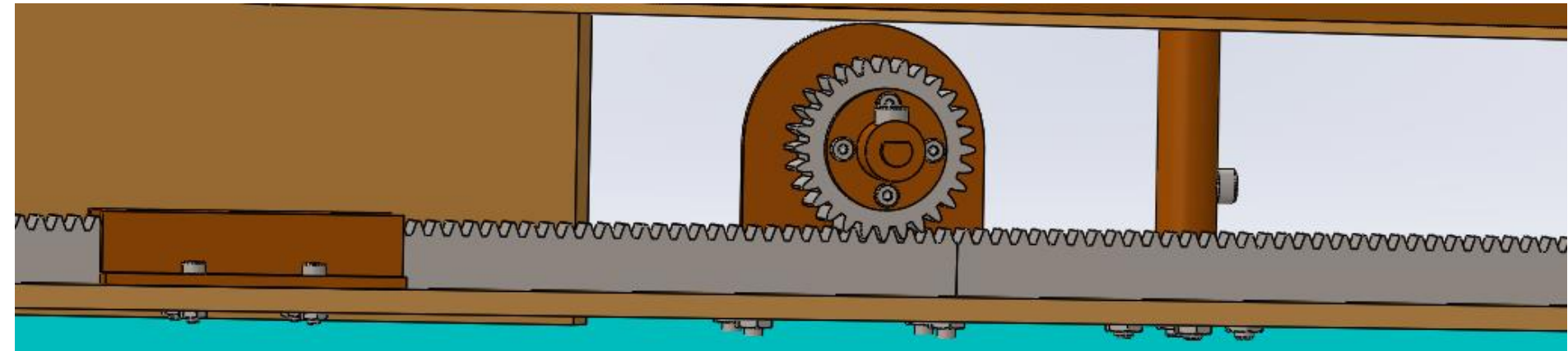


Figure 7: The system's horizontal arm reaches out through the use of a rack and pinion system.

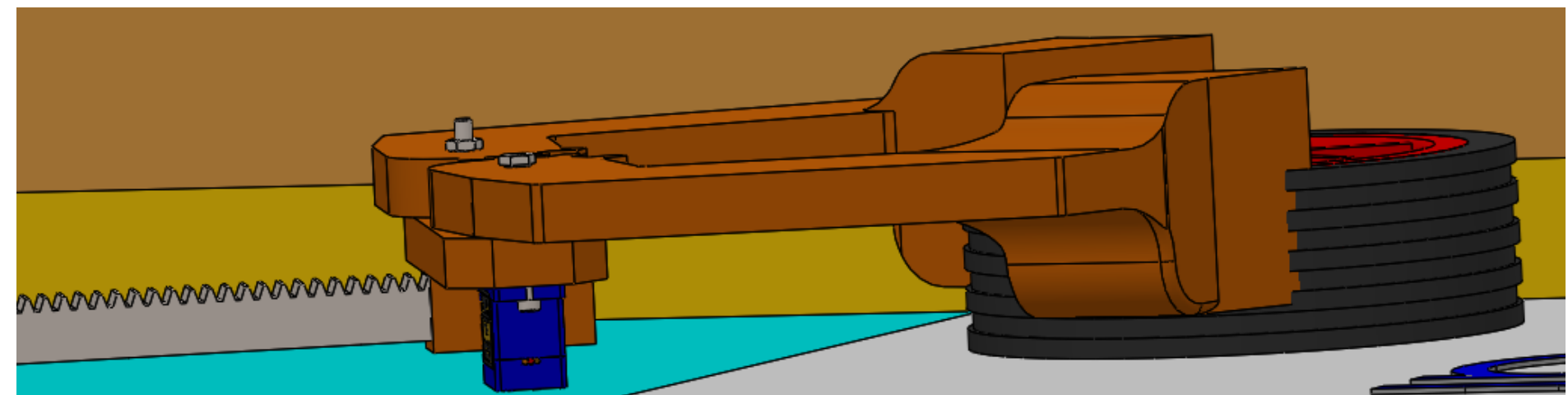


Figure 8: The system will use a claw mechanism powered by a servo motor to clasp onto the package.

2.1 CAD Images (Traversing Chasm)

PHASE 3

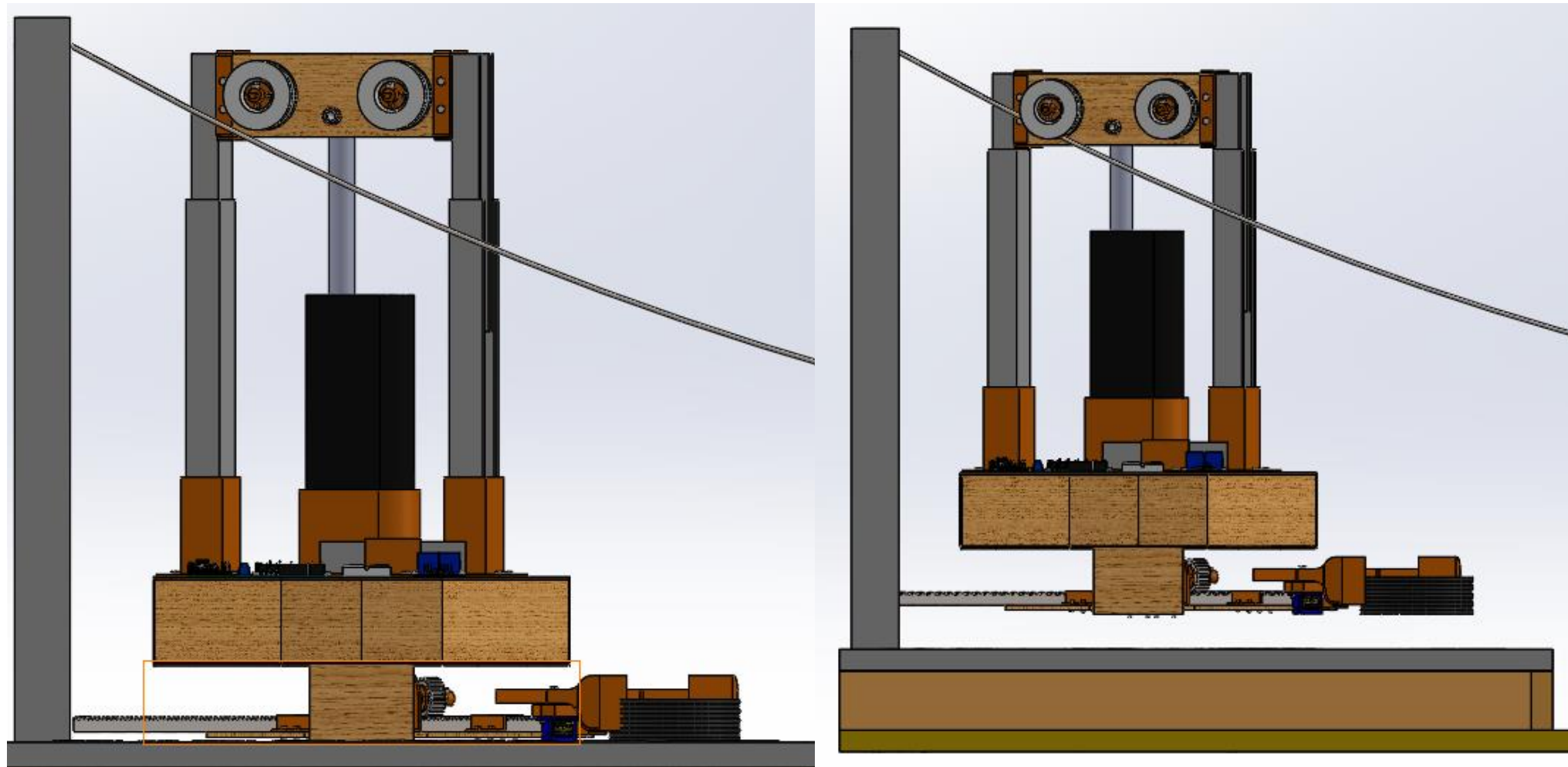


Figure 9: Once the system has the package, the linear actuator will retract until the system is solely dependent on the wire.

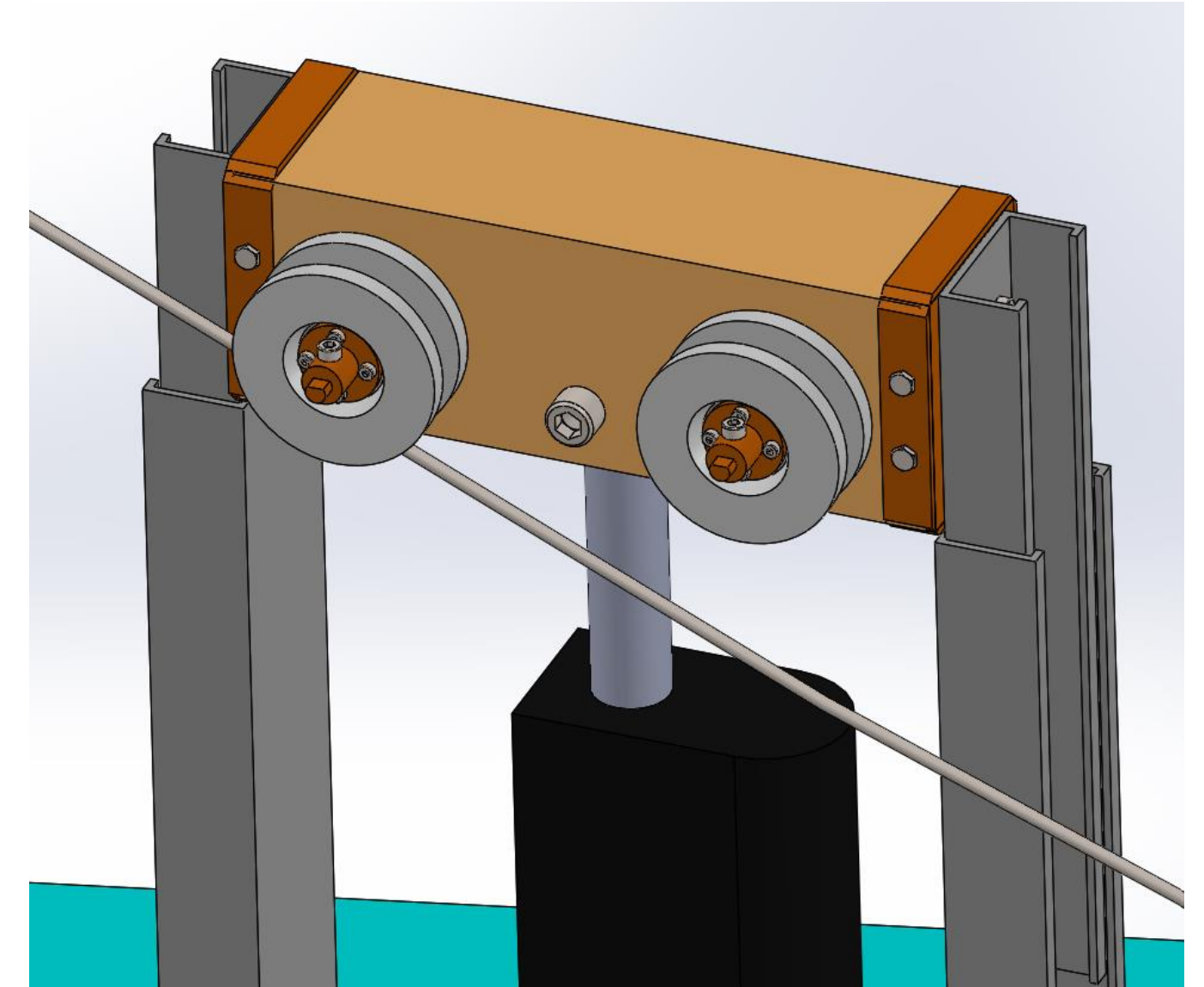


Figure 10: The system uses two motorised pulleys to traverse across the wire.

2.1 CAD Images (Traversing Chasm)

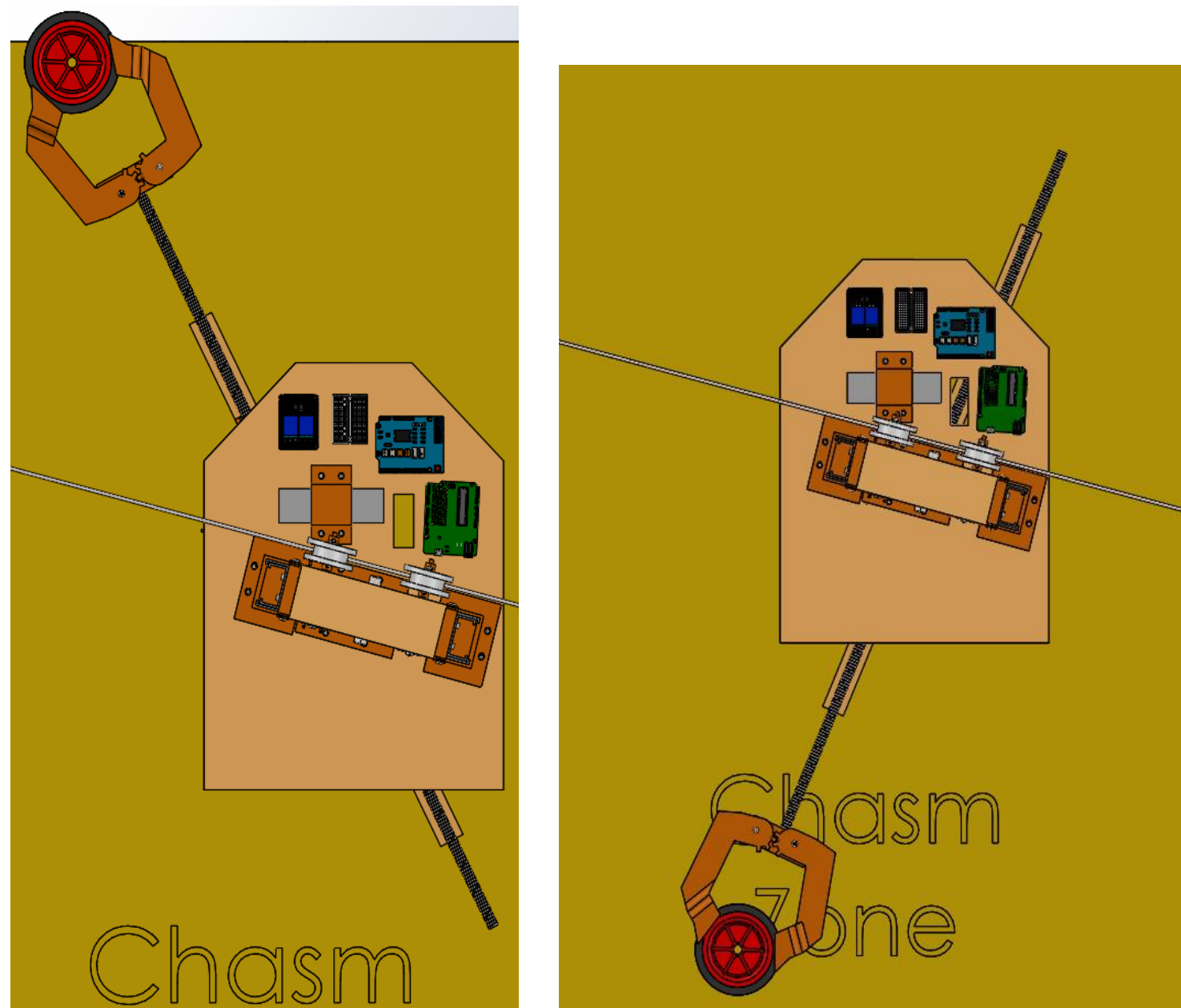


Figure 11: While traversing across the Chasm, the arm will rotate the package so that it is in position and ready for depositing. It needs to rotate in the Chasm as there is not enough space at the Deposit Zone for the rack to turn without hitting the pylon.

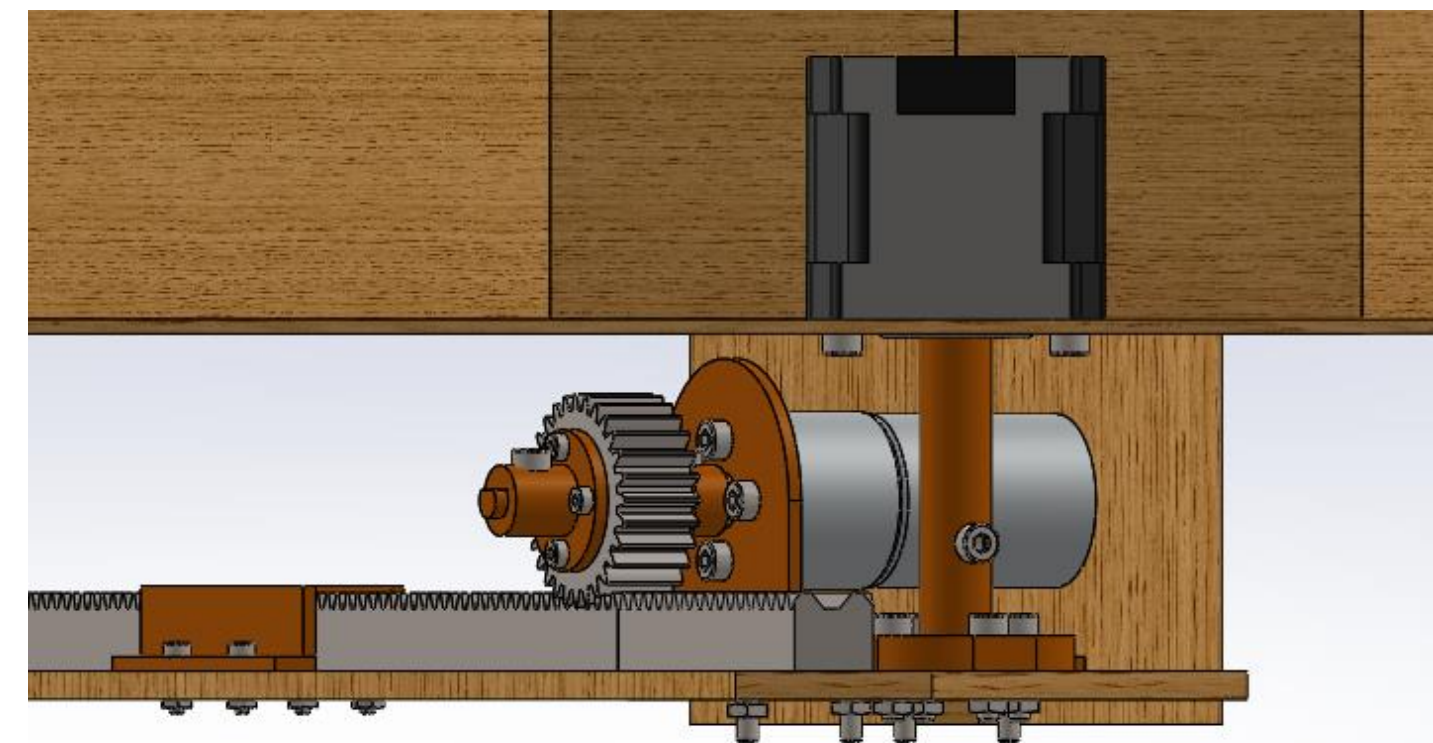


Figure 12: A stepper motor and a designed mounting hub that connects to the base plate are used to rotate the system.

2.1 CAD Images (Depositing Payload)

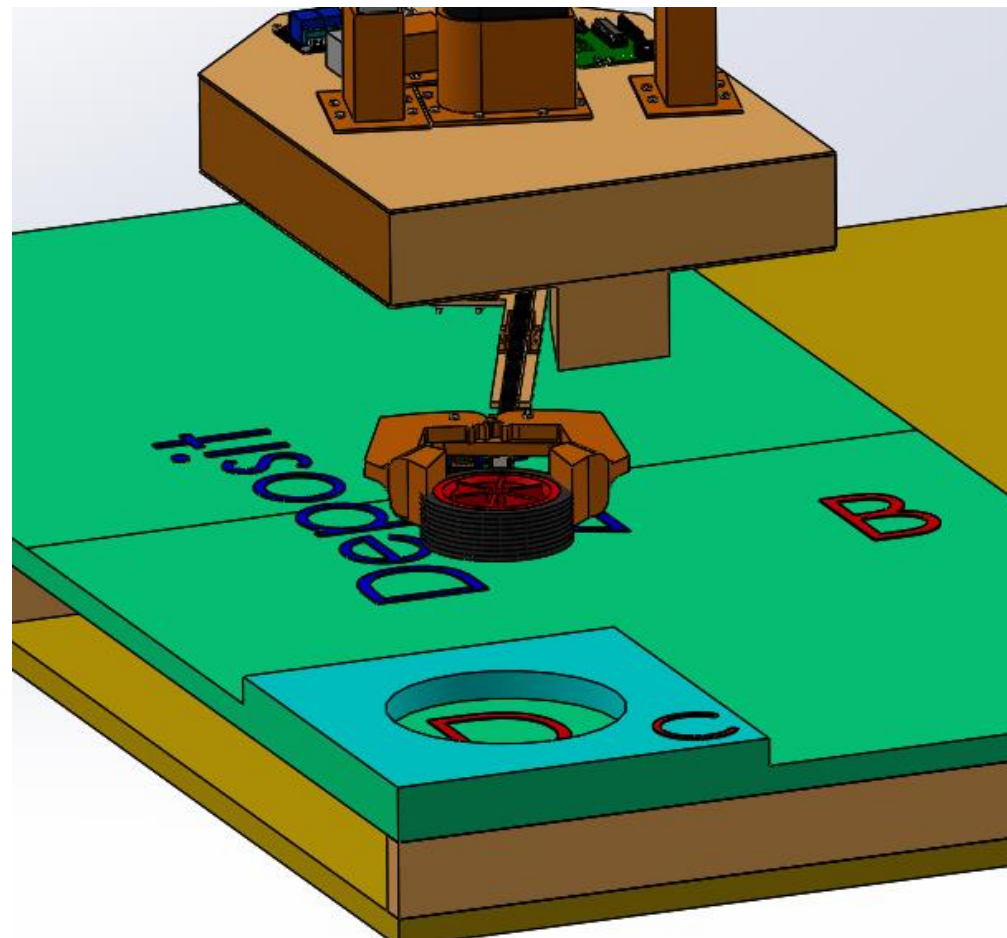


Figure 13: System arrived to the Deposit Zone and is aligned with Deposit Zone D.

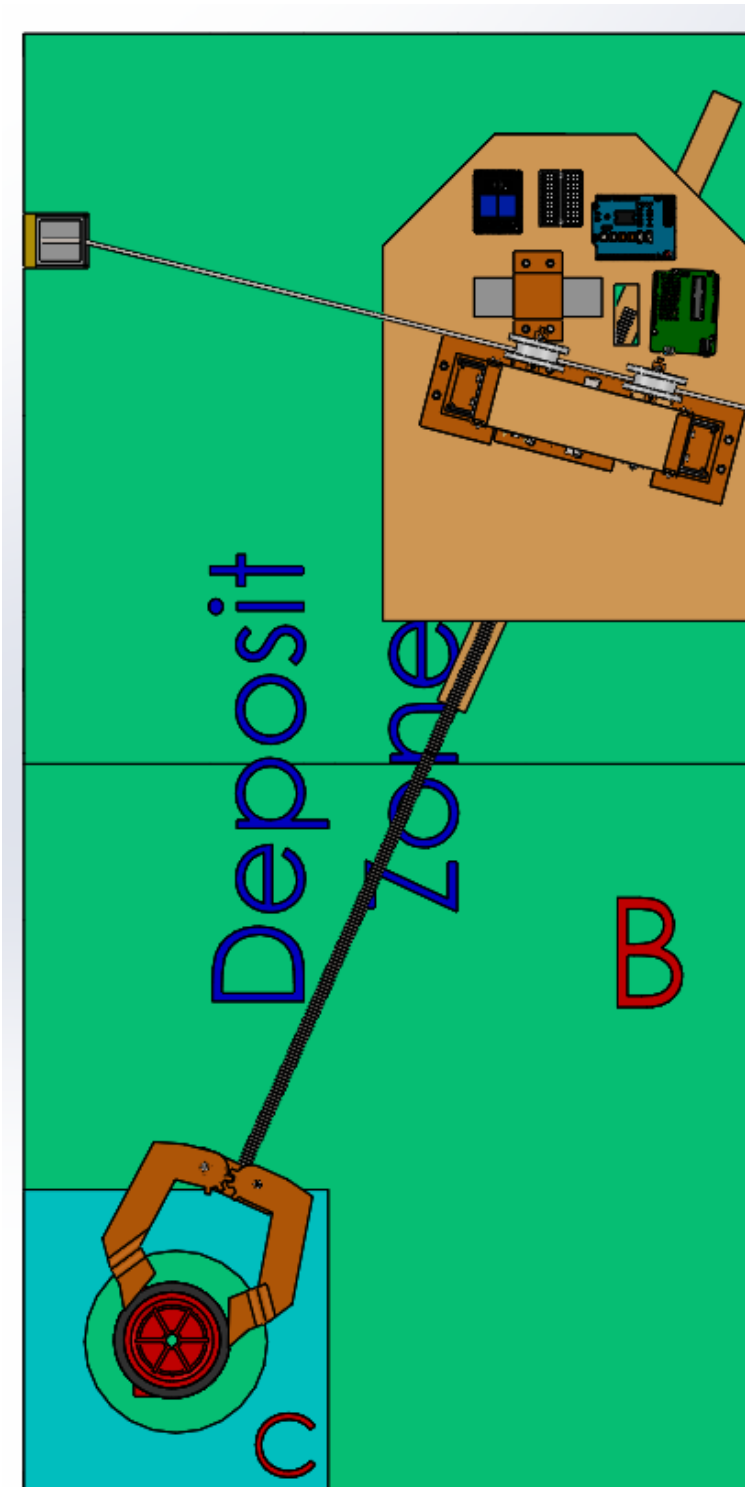


Figure 14: Horizontal arm extending to drop the package at Zone D.

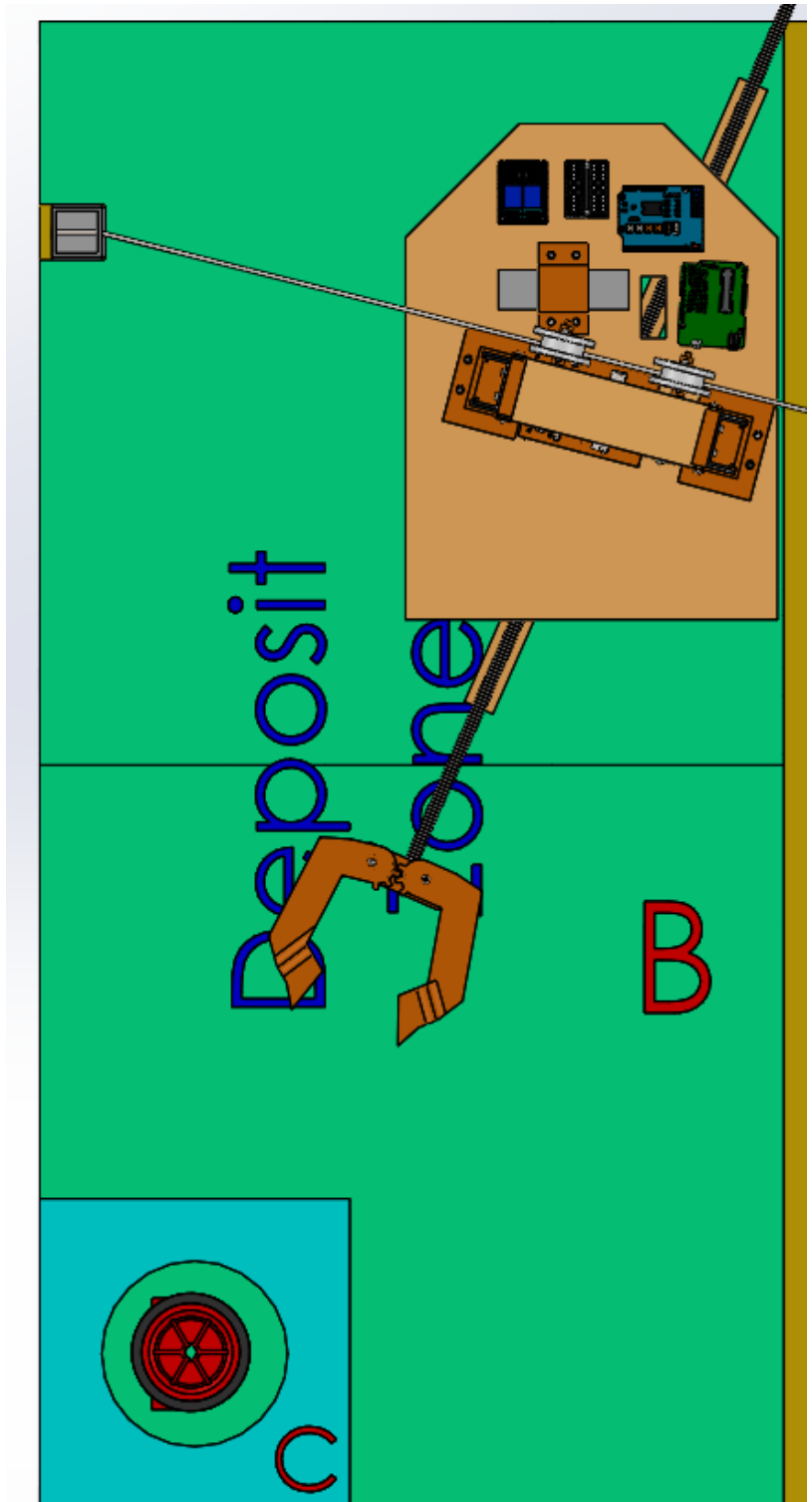


Figure 15: Horizontal arm retracting.

2.1 CAD Images (Returned to Start Zone)

PHASE 5

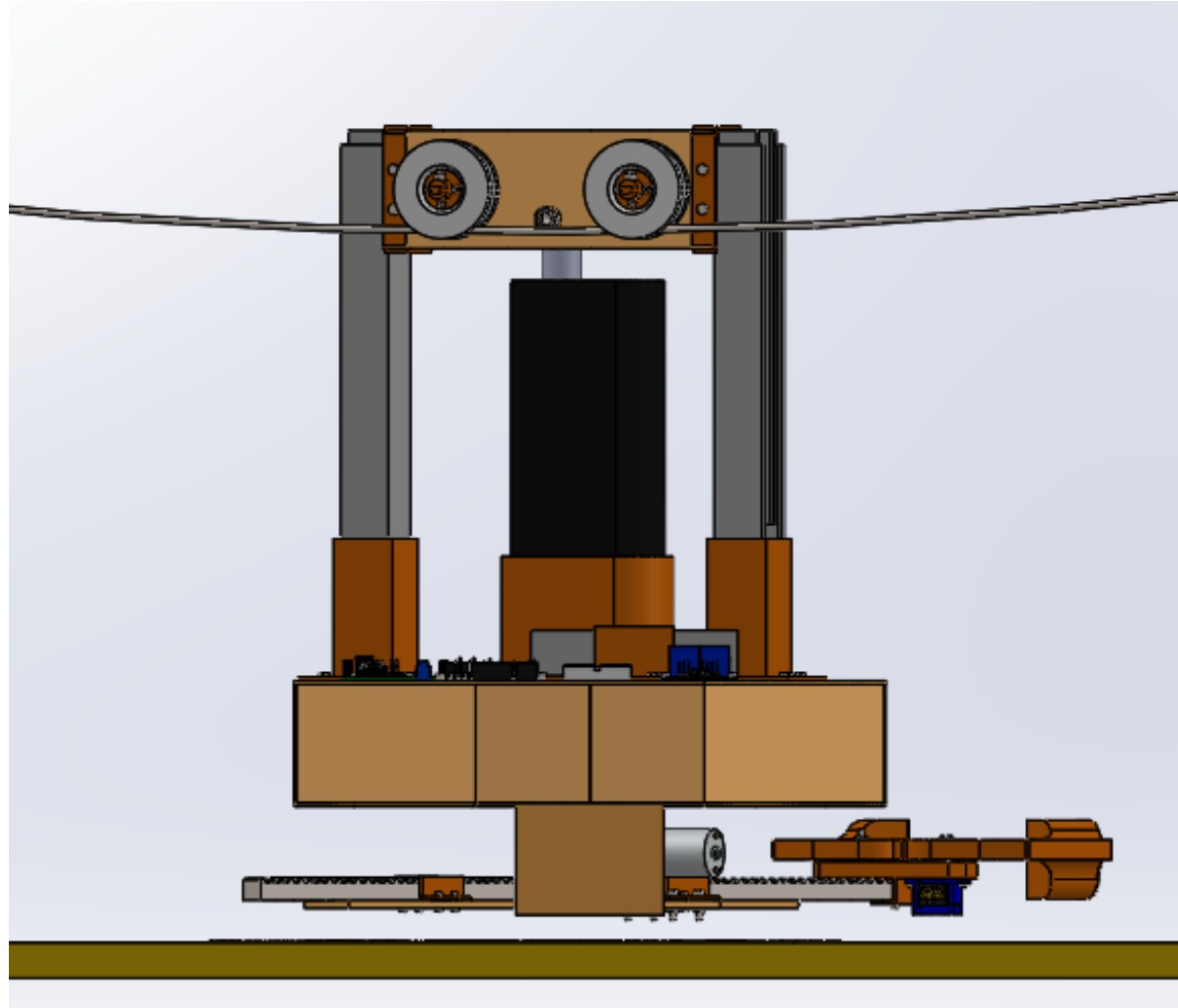


Figure 16: The system utilises the pulleys to traverse back to the Start Zone. At the sag point, the system is light enough to not touch the floor.

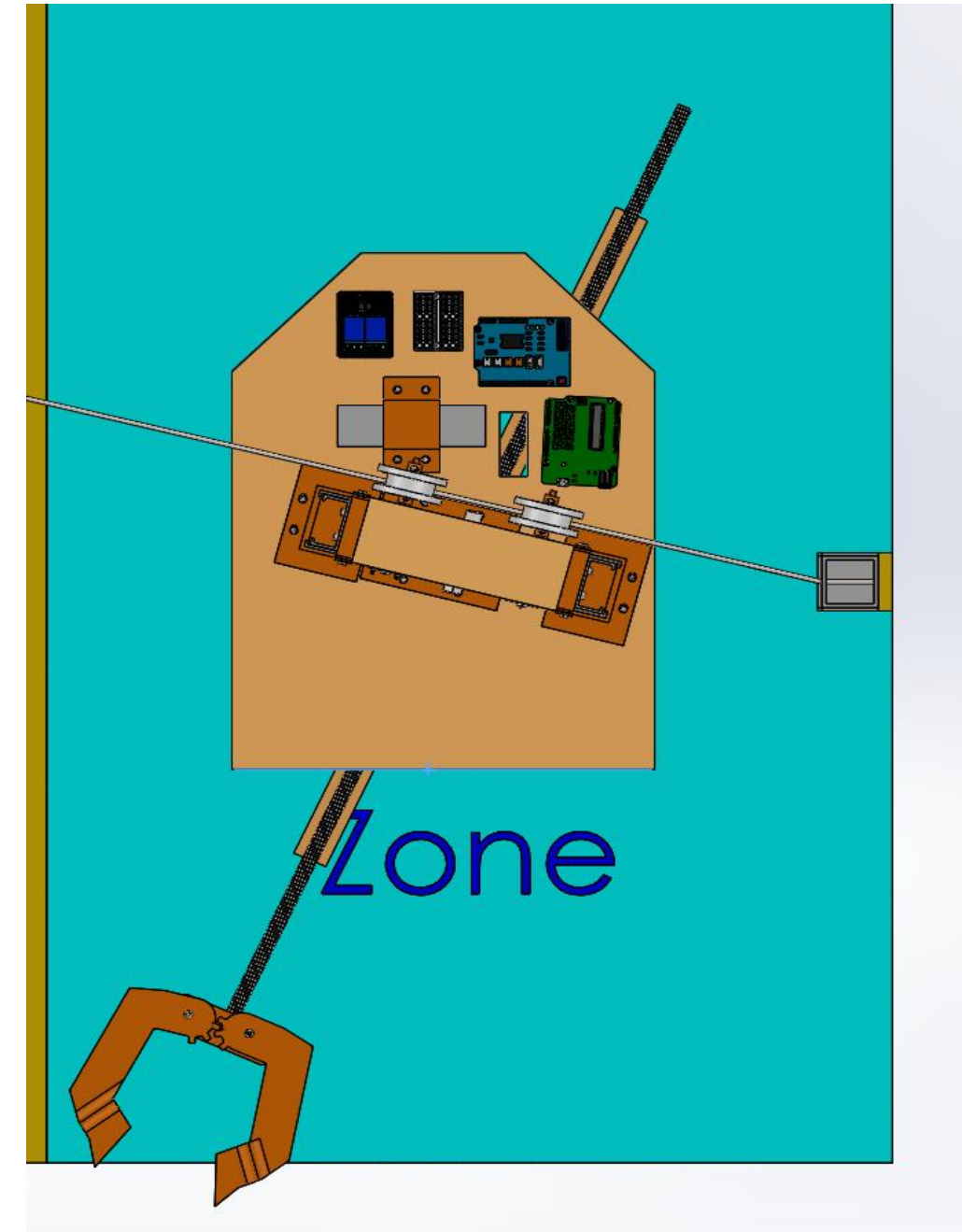


Figure 17: The system returns back into the Start Zone. It come in with far enough so that the arm is within the bounds of the Start Zone.

2.2 Final Manufactured Device Images (Set-up and ready for start) **PHASE 1**

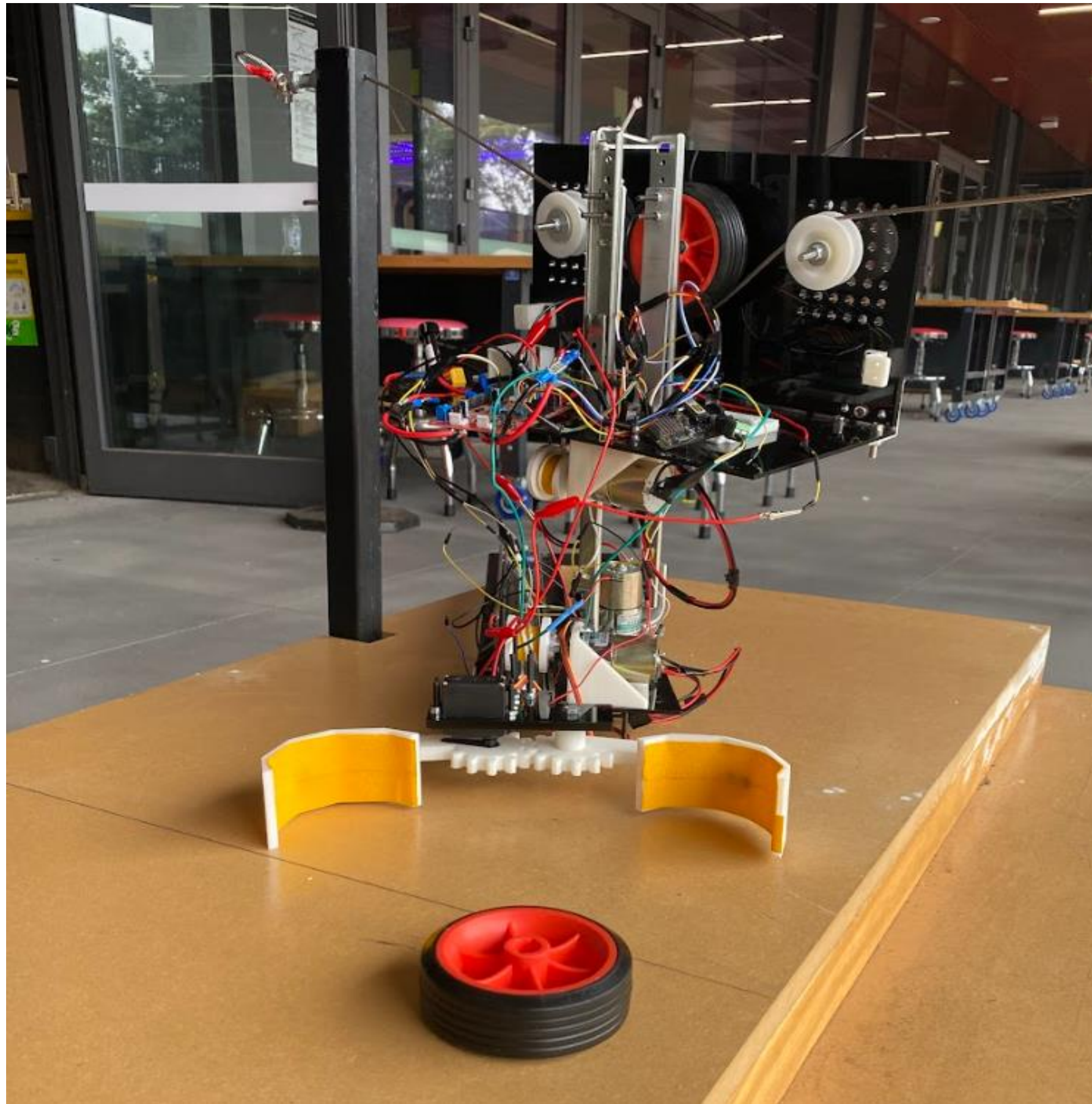


Figure 18: System is set up on the wire, with the claws aligned with the package.

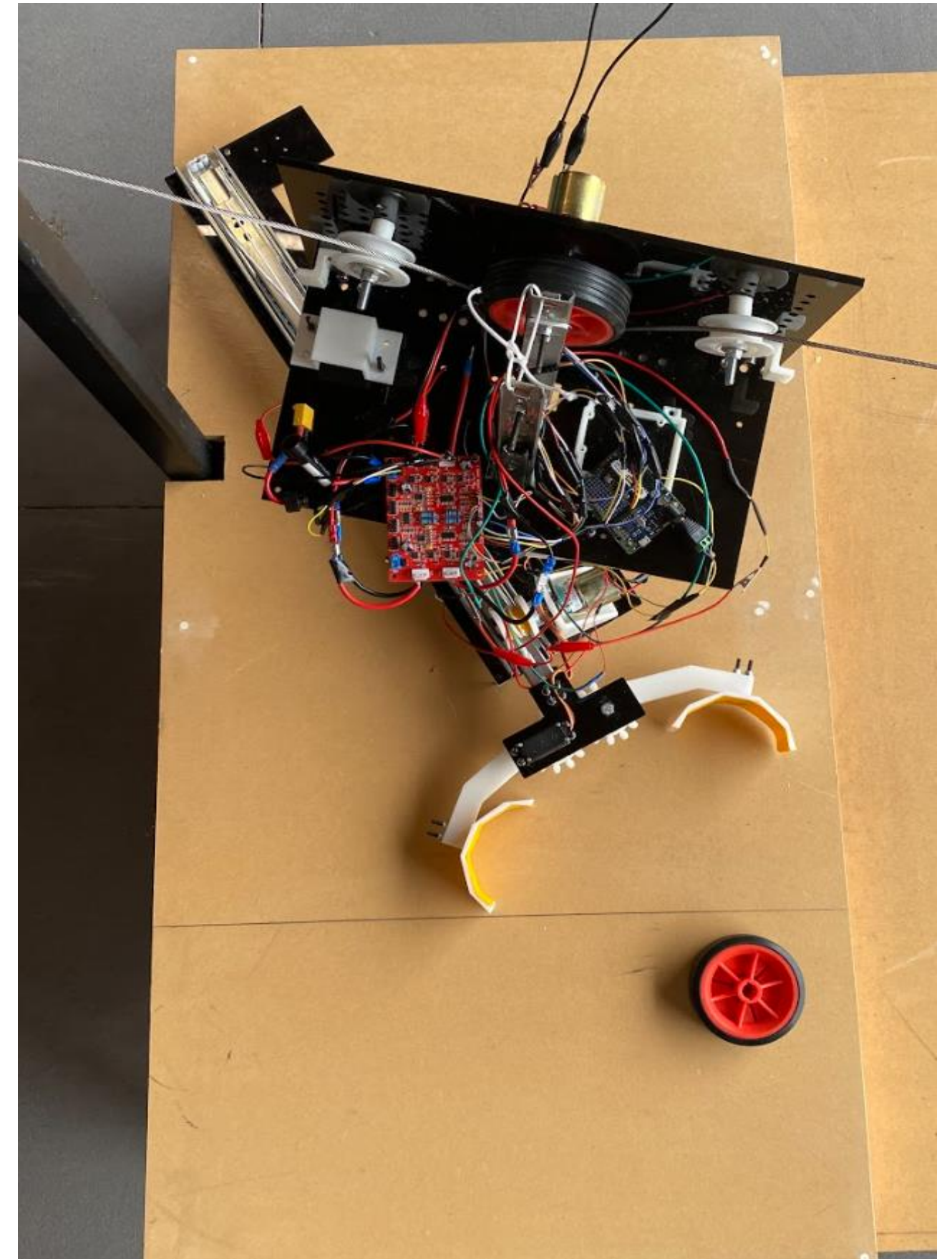


Figure 19: The system fits perfectly within the Start Zone.

2.2 Final Manufactured Device Images (Collecting Payload)

PHASE 2

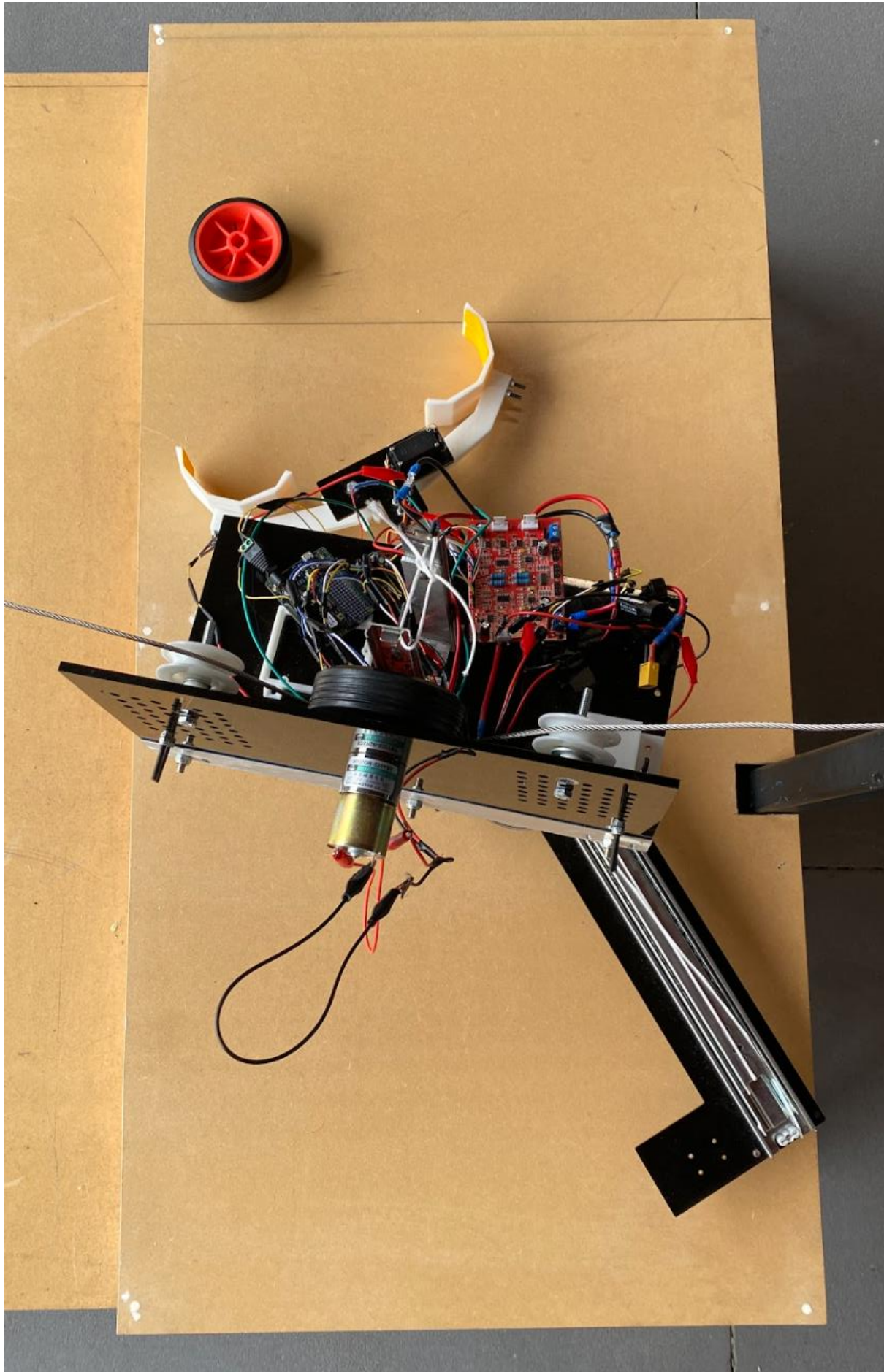


Figure 20: The system before it collects the payload.

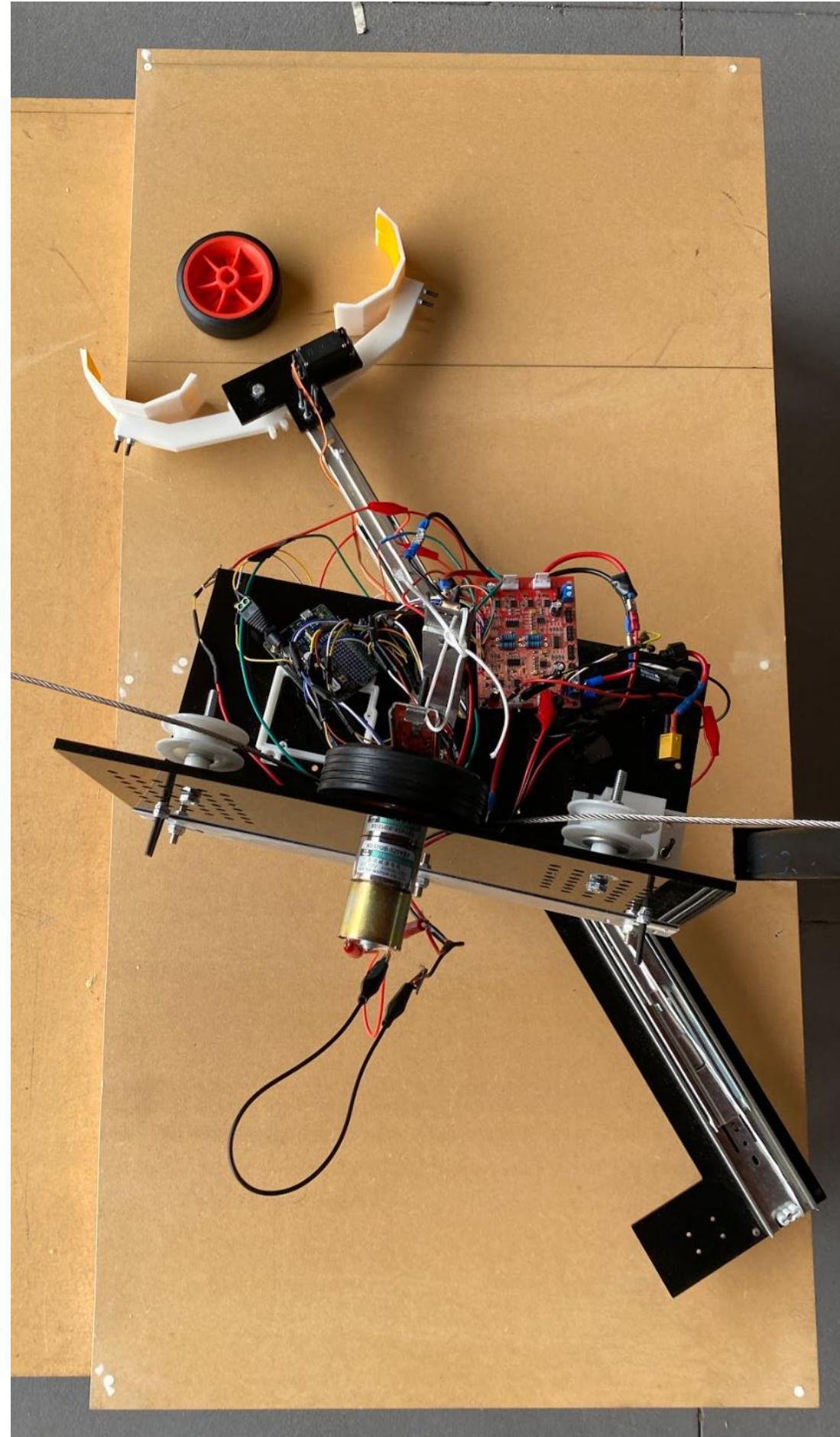


Figure 21: The system is timed and actuates out far enough to be in position of the payload.

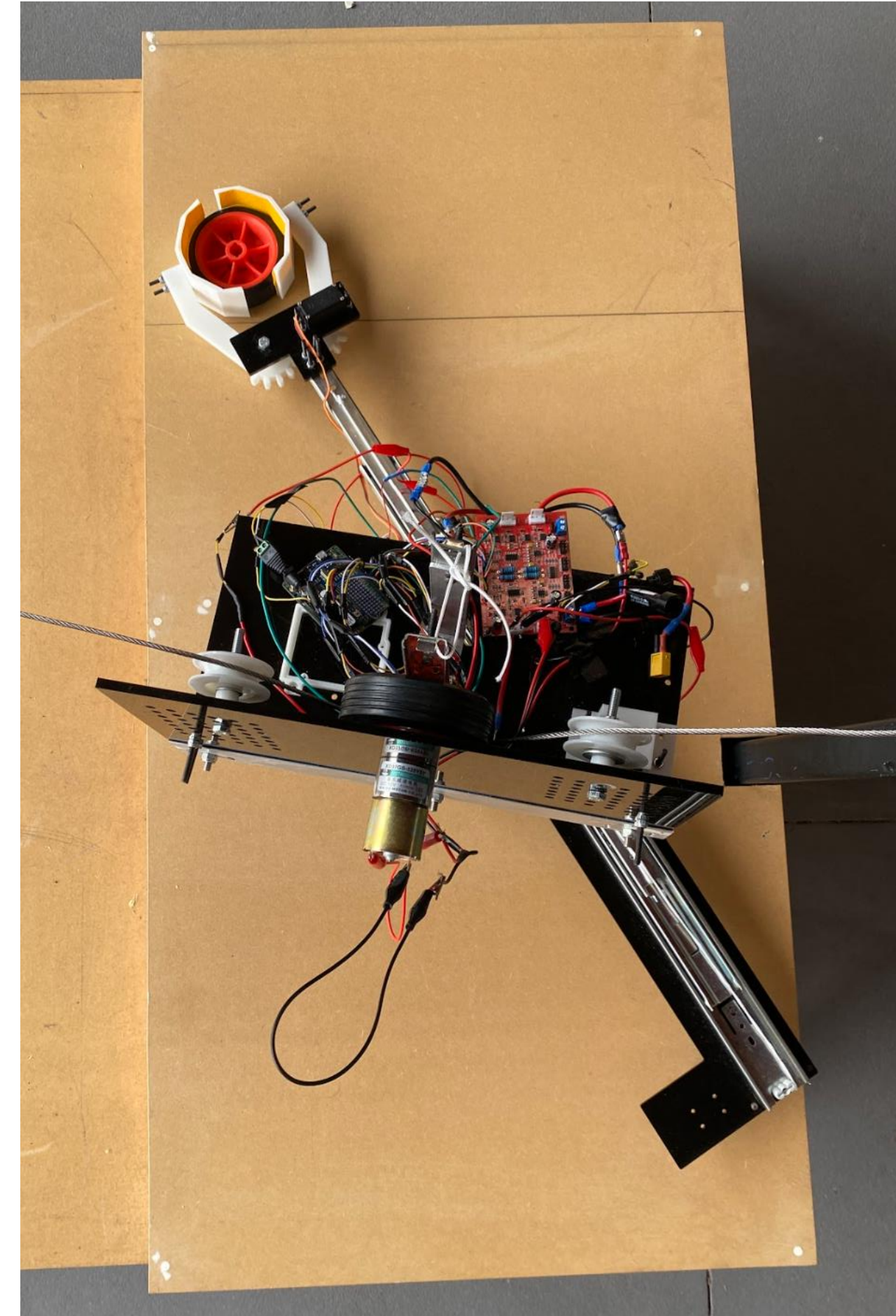


Figure 22: The claws close onto the payload, collecting it.

2.2 Final Manufactured Device Images (Collecting Payload)

PHASE 2

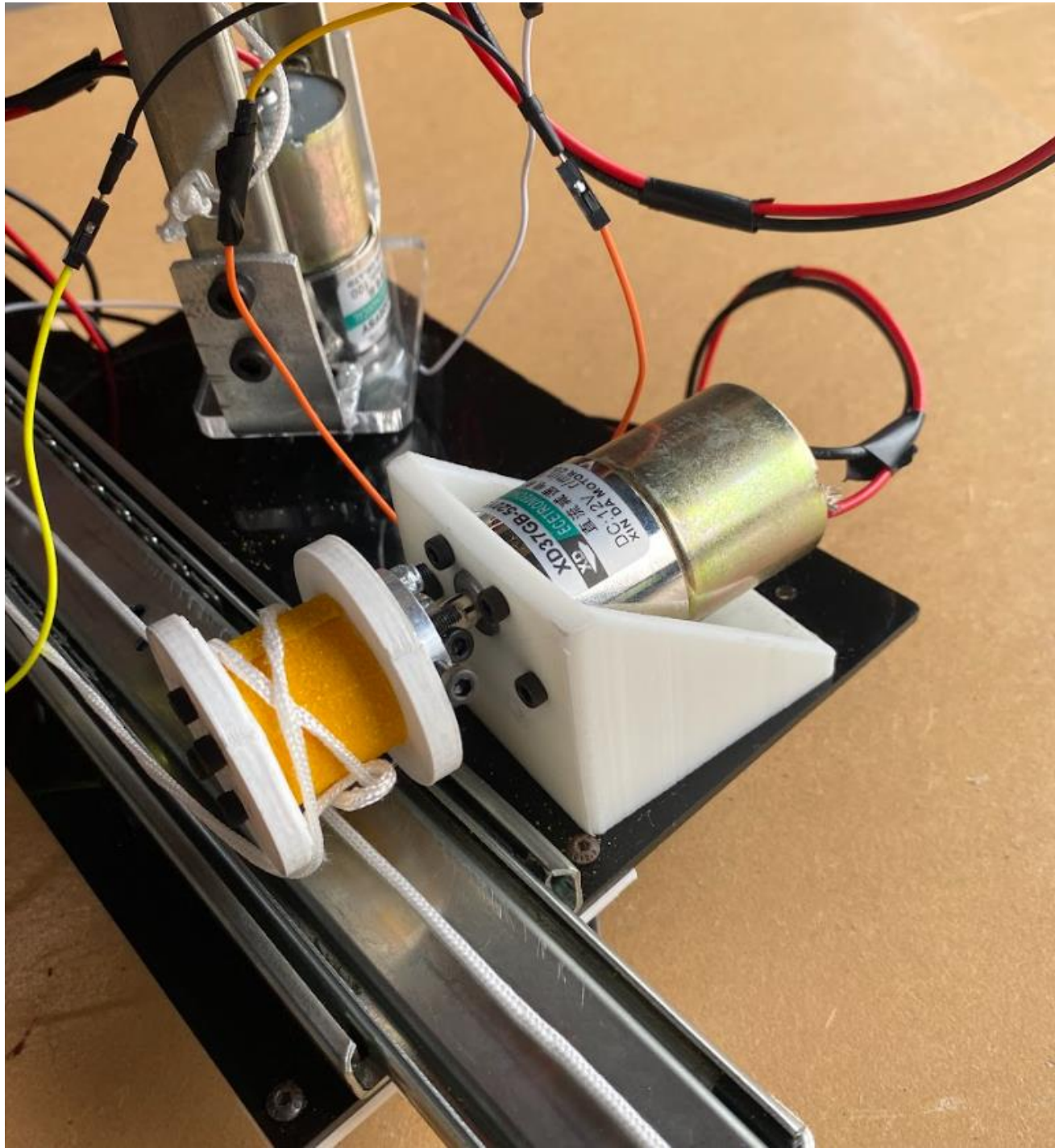


Figure 23: To actuate the drawer slides out, the system uses a spool and drawer slide mechanism.

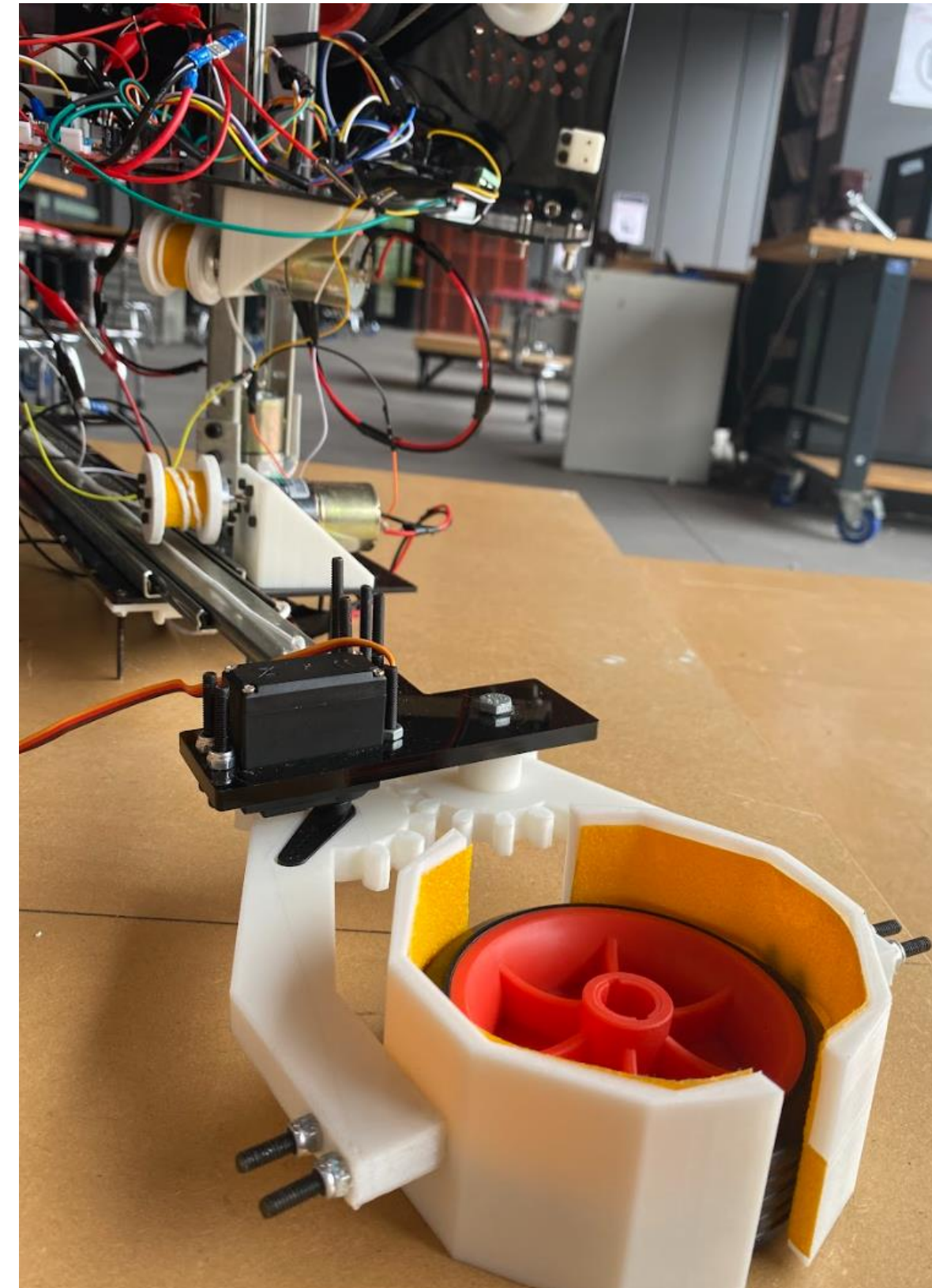


Figure 24: To clasp the payload, the system utilizes a servo motor and geared teeth.

2.2 Final Manufactured Device Images (Traversing Chasm)

PHASE 3

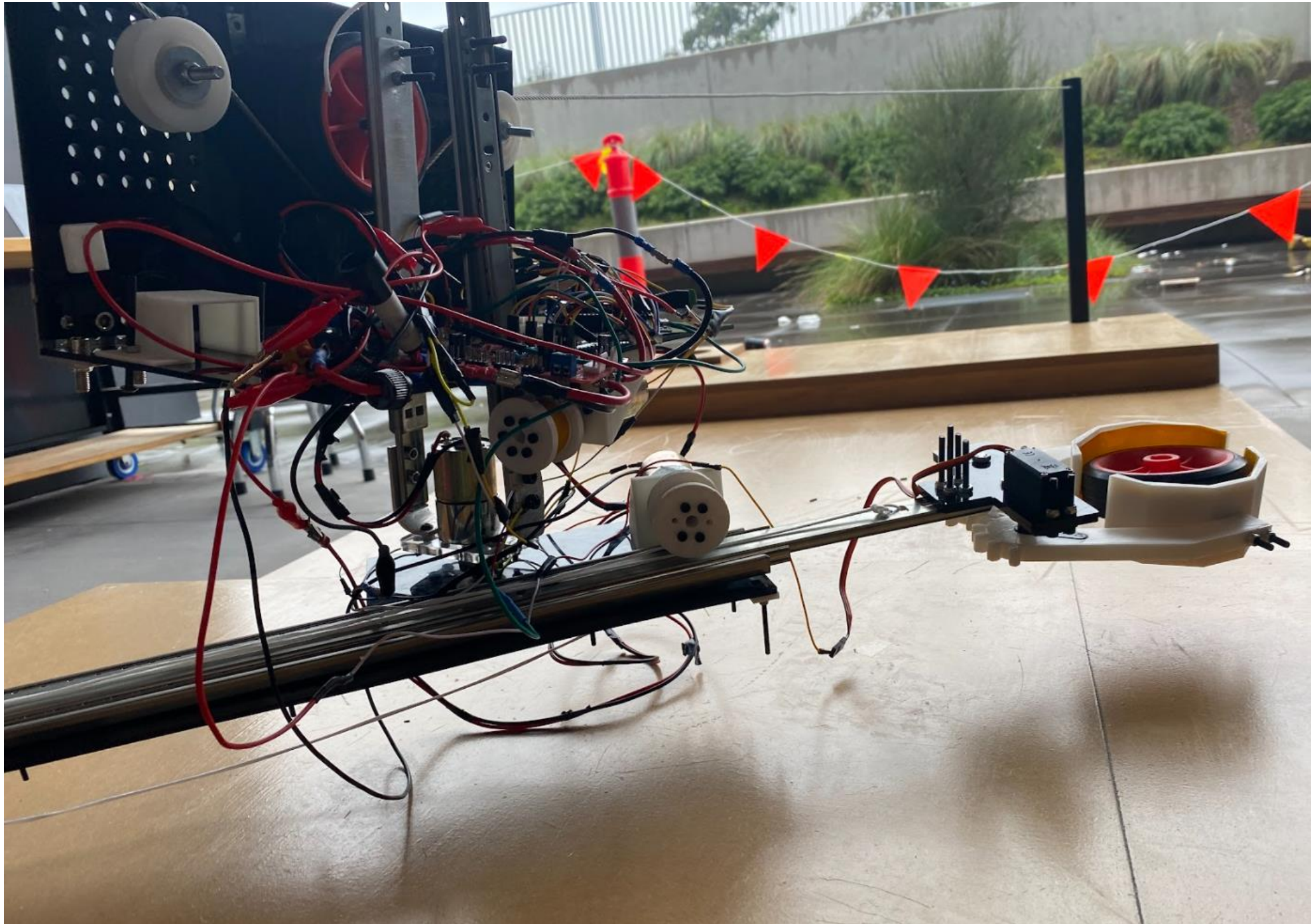


Figure 25: To traverse the wire, the system needs to first lift.

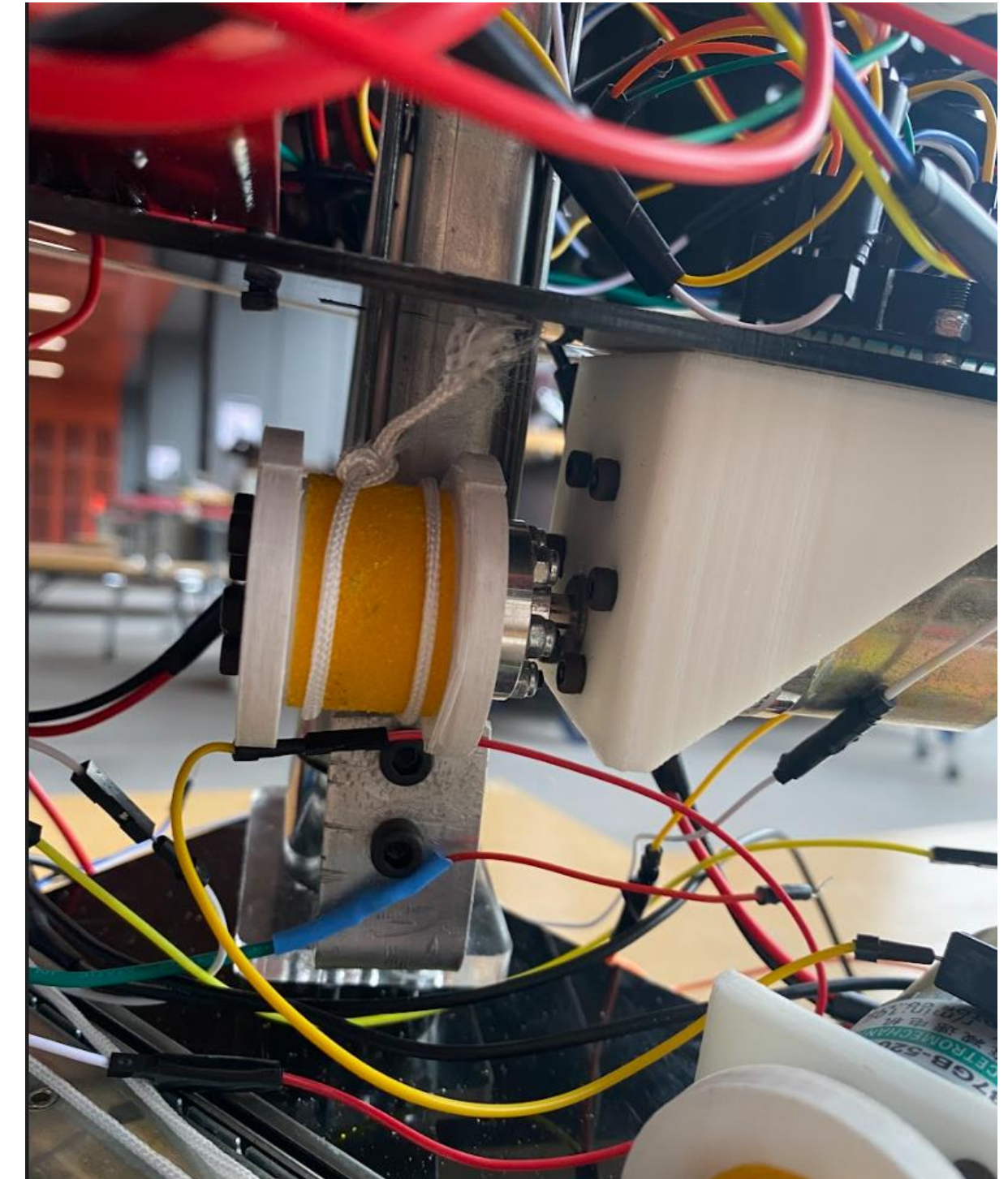


Figure 26: The system lifts up using a spool and drawer slide mechanism.

2.2 Final Manufactured Device Images (Traversing Chasm)



Figure 27: To system traverses the wire using one drive wheel and two pulleys on either side as guides.

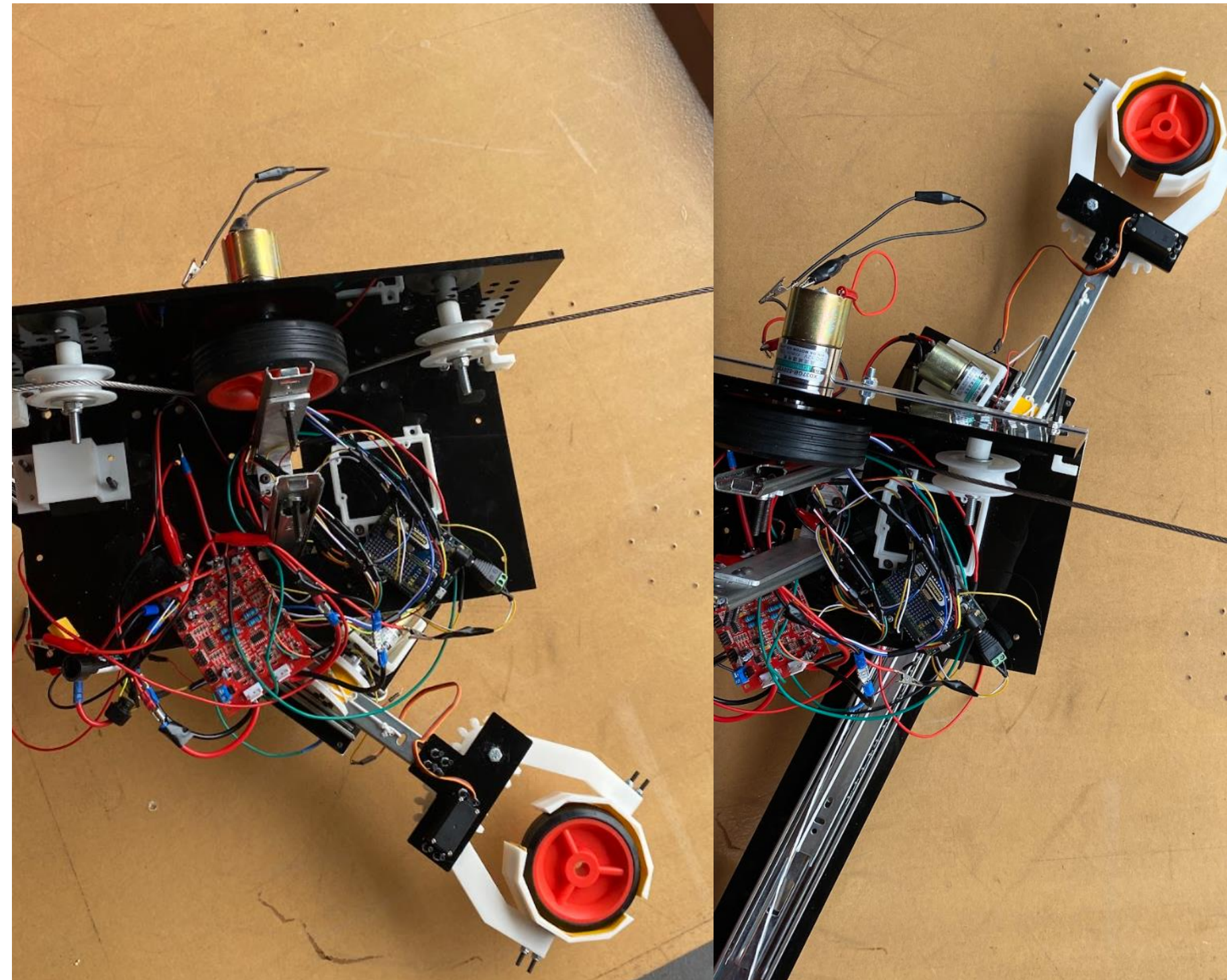


Figure 28: While traversing the wire, the system rotates the arm to set it up for the deposition. The image on the left depicts the arm before rotation, the image on the right depicts the arm after rotation.

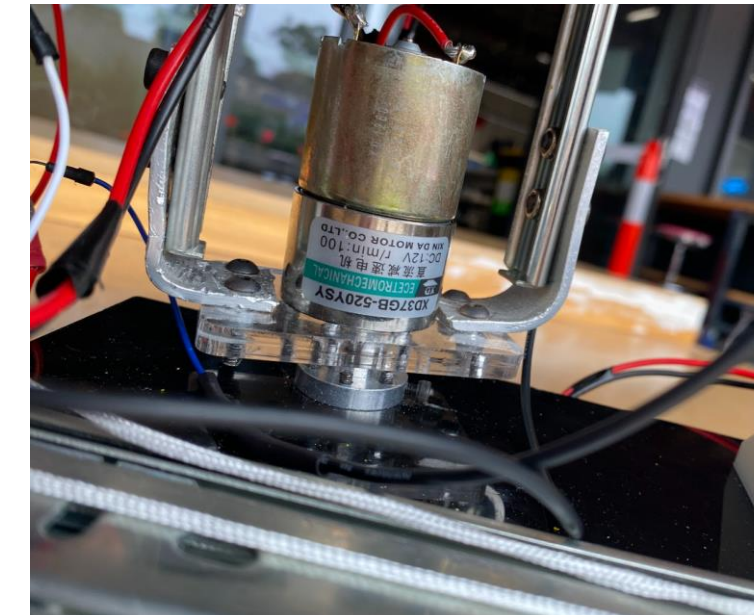


Figure 29: The system utilises a DC motor connected to the base rotating plate via a mounting hub to rotate the arm.

2.2 Final Manufactured Device Images (Depositing Payload)

PHASE 4

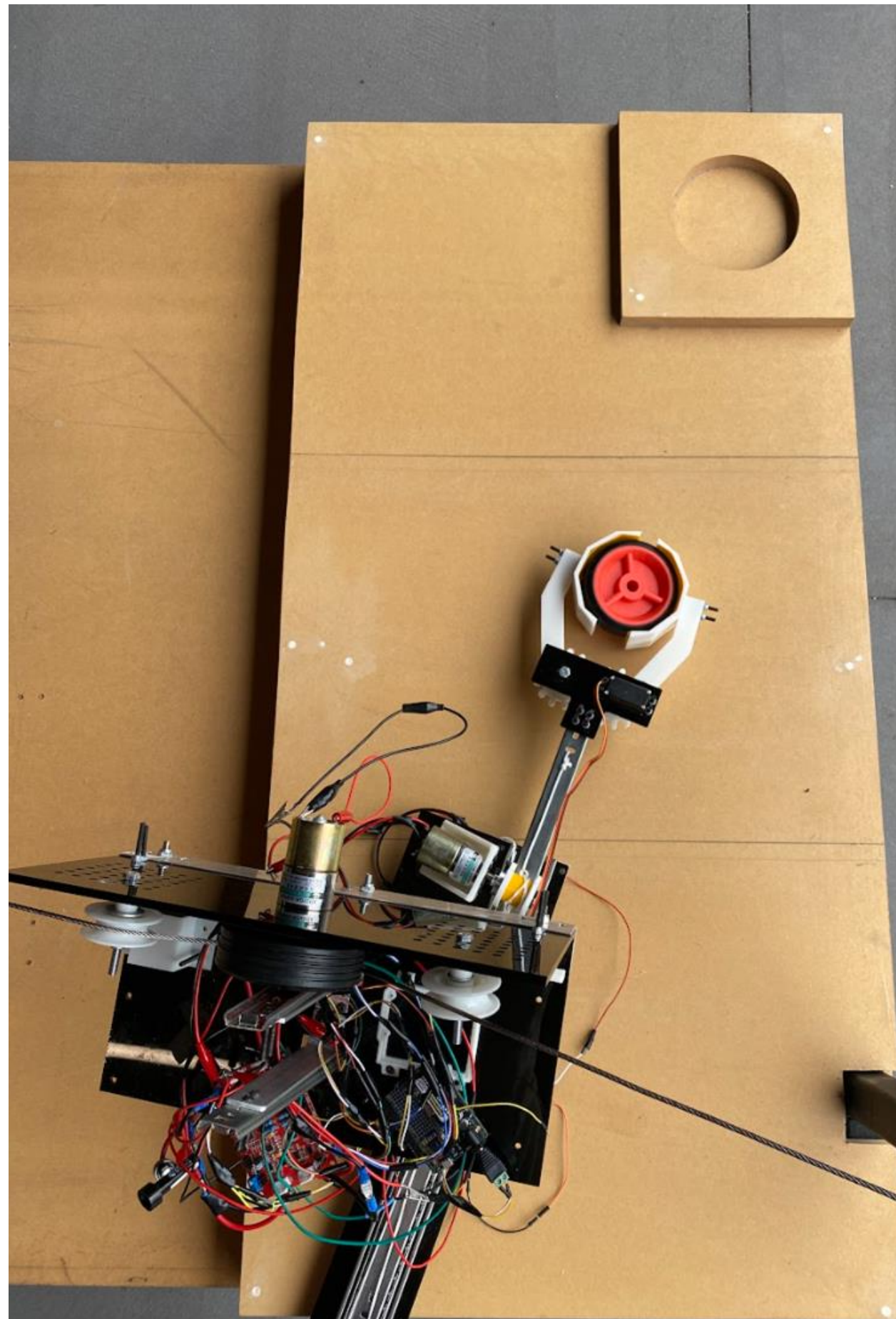


Figure 30: The system before it deposits the payload.

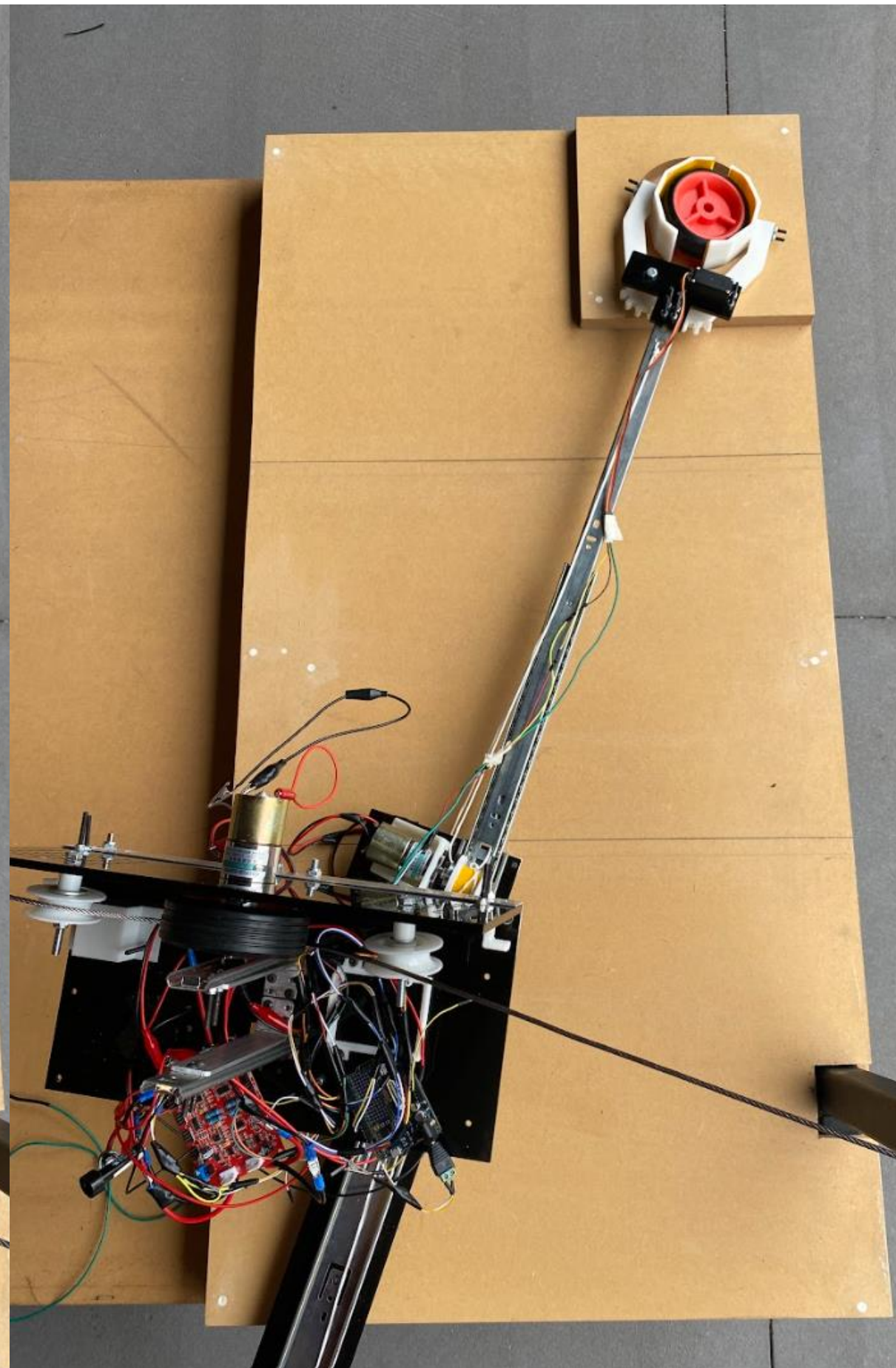


Figure 31: The system is timed and actuates out far enough to be in Deposit Zone D.

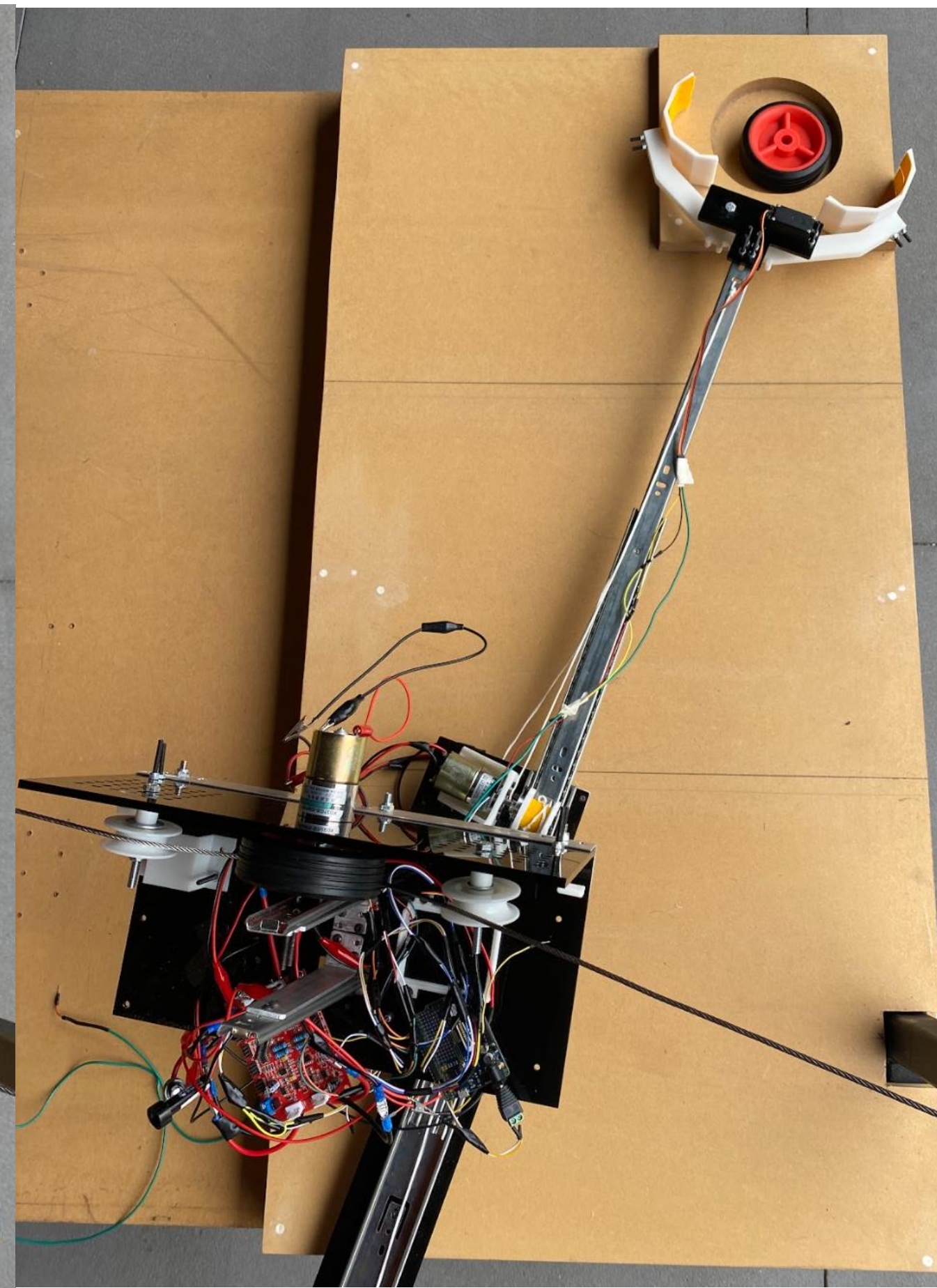


Figure 32: The claws open, dropping the payload in Deposit Zone D.

2.2 Final Manufactured Device Images (Returned to Start Zone) **PHASE 5**

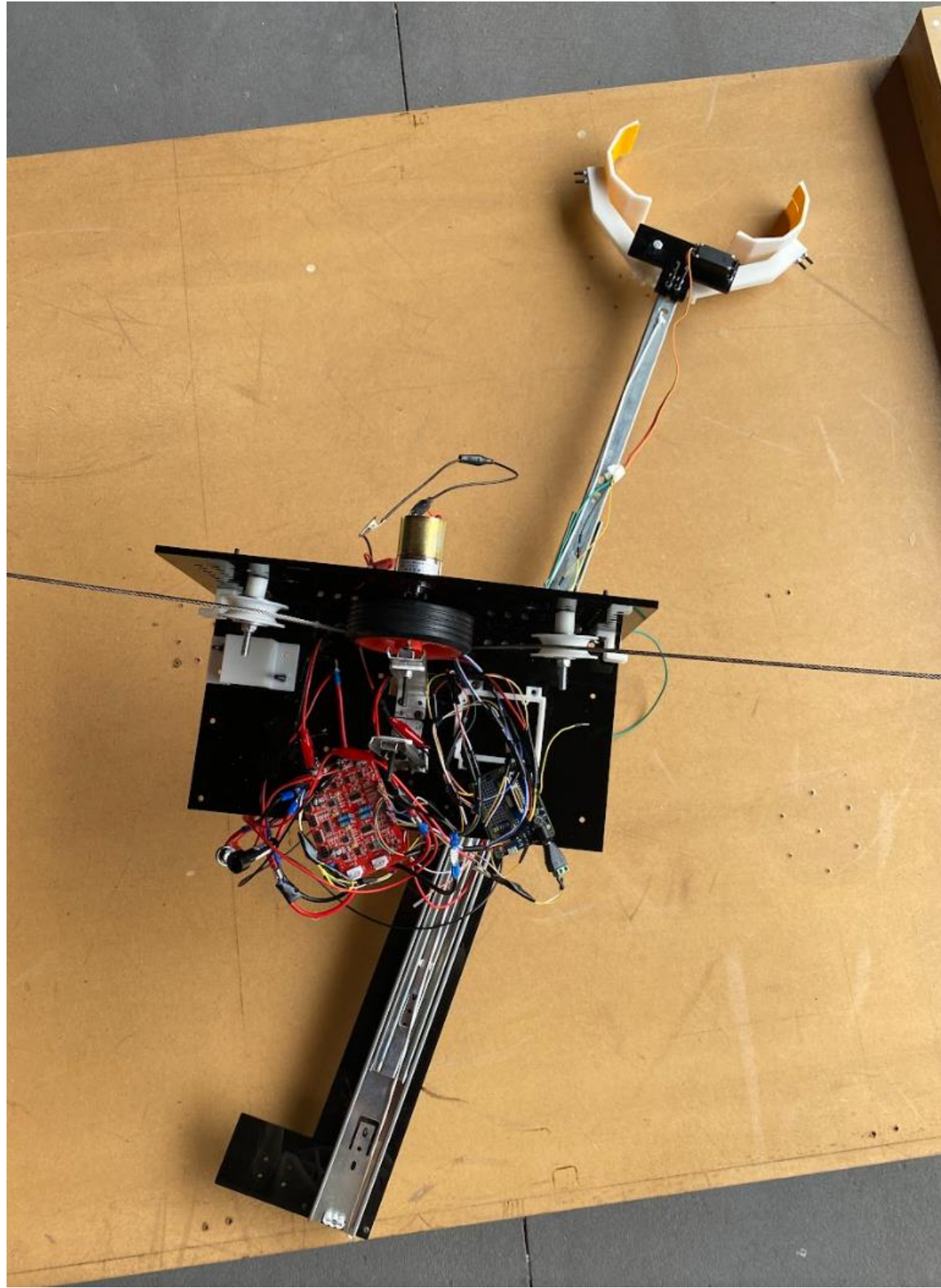


Figure 33: The system returns back to the Start Zone using the drive wheel and pulleys.

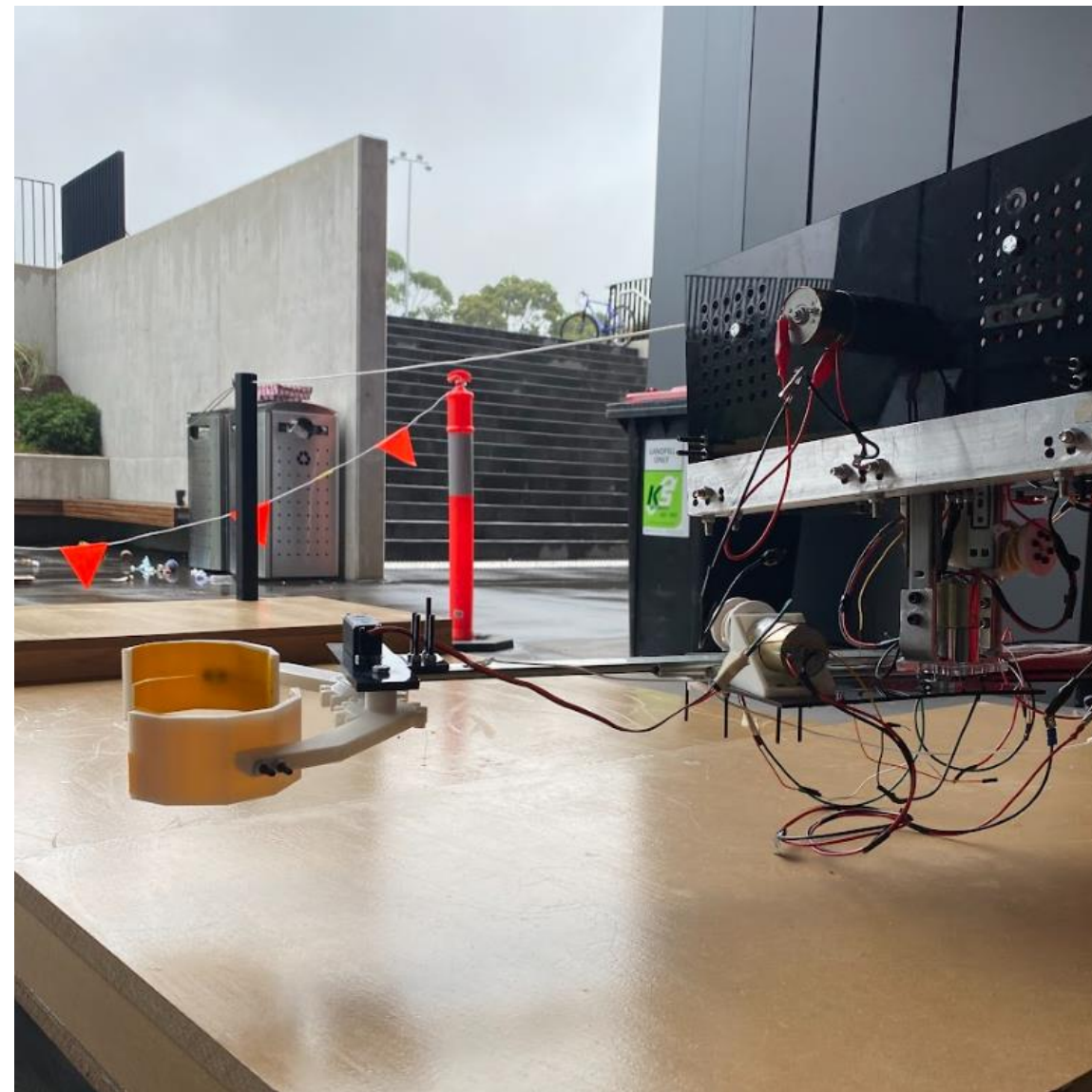


Figure 34: The system retracts the arm slightly and remains elevated when done.

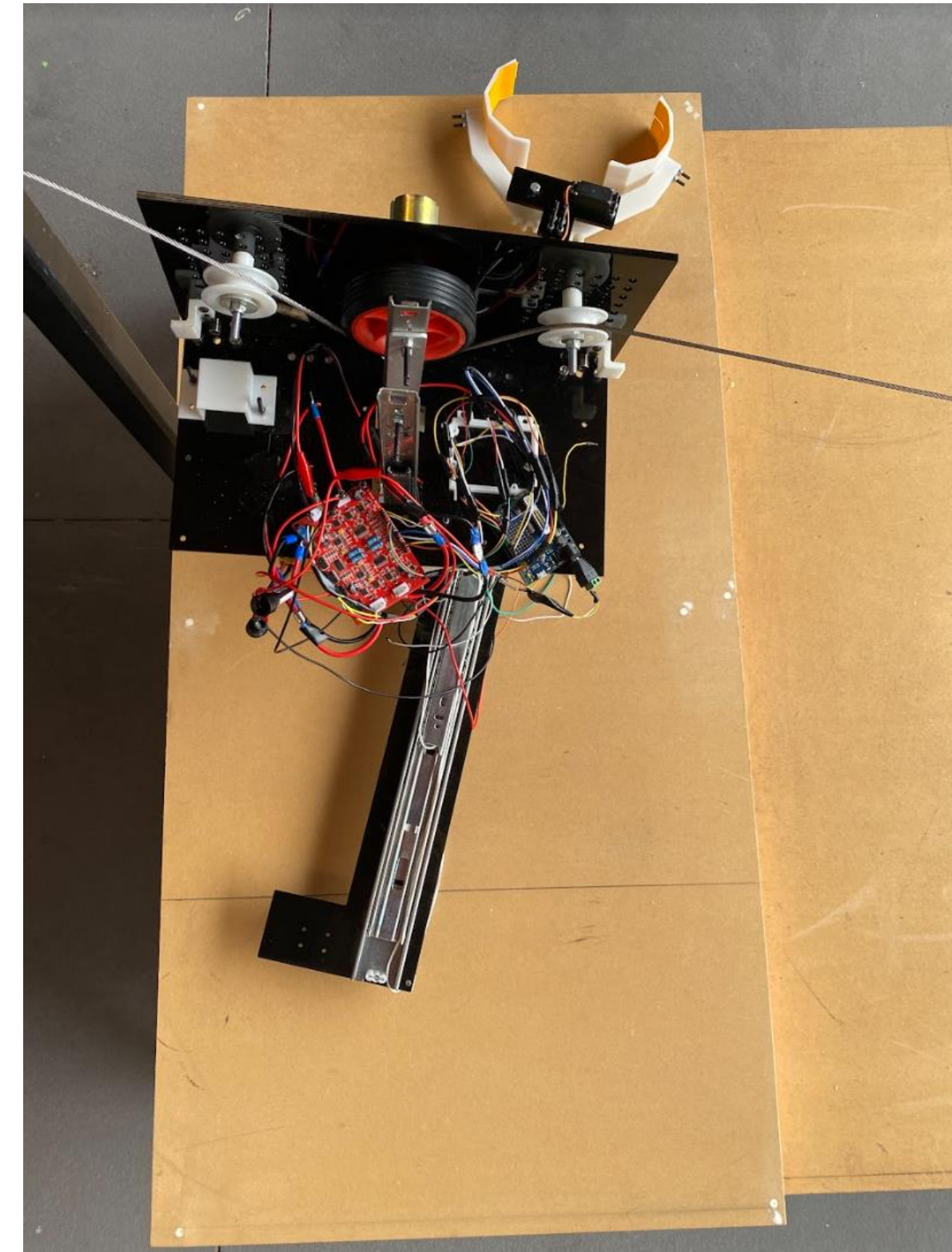


Figure 35: The system fits comfortably in the Start Zone.

2.3 Design Comparison - First Major Difference (Word Count: 415)

The first major difference between our Preliminary CAD Concept and our final manufactured device is that our preliminary concept utilised a linear actuator to raise and lower the system, while the final manufactured device utilised a spool and drawer slide mechanism to lower and raise the system. In the preliminary submission, the team designed the system to use a linear actuator that will actuate upwards with drawer slides as support on the sides as illustrated in Figure 9. However, this mechanism changed and improved to become a spool and drawer slide mechanism to actuate up as can be seen in Figure 26. Originally in the preliminary submission, the team assumed we could not start on the wire and as such we designed the system to use a linear actuator to actuate the wire traversal mechanism up and hook onto the wire. However, due to the rules being clarified to us after the preliminary CAD, we discovered teams are allowed to start on the wire, and as such, our team believed it was best to start on the wire as it would reduce the complexity of our system’s mechanism. Furthermore, after investigating costs, the team discovered that most linear actuators that are quickly accessible and available to buy are very expensive and it would blow our budget up higher than forecasted. As a result, the team thought of utilising a spool and drawer slide mechanism to actuate our system up since it would only require a DC motor which are often much cheaper than linear actuators. Hence, with our new design idea, the team did a new CAD of the system, illustrated in Figure 36. CAD was helpful in trying to incorporate ideas. However, testing was even more revealing. The team bought necessary motors and drawer slides and mounted it underneath our system to follow the design of the new CAD, as illustrated in Figure 37. However, thankful due to testing, the team discovered that mounting the drawer slides underneath the wire traversal plate meant the mechanism would be too tall when in the Chasm zone and it could not actuate up any higher (see Figure 37) . Consequently, testing lead to the refinement of our idea, and as such we figured a solution that had the drawer slides sticking through the top plate - indicated in Figure 38, to allow for the system to actuate up high enough to traverse the Chasm. Will all problems now resolved, this was our final manufactured vertical actuation mechanism.

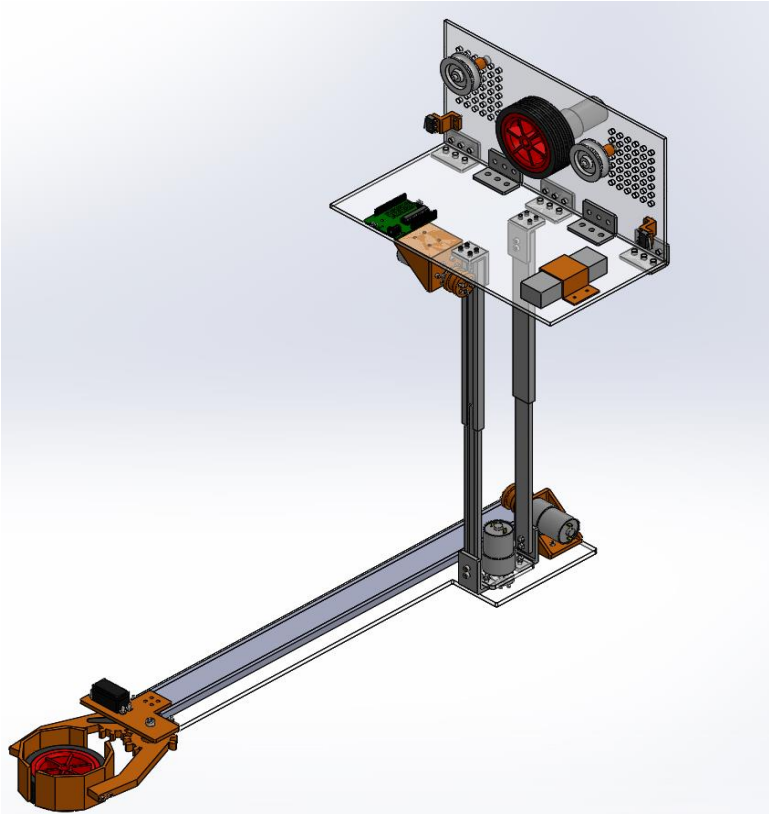


Figure 36: CAD of new system after receiving clarity on rules.

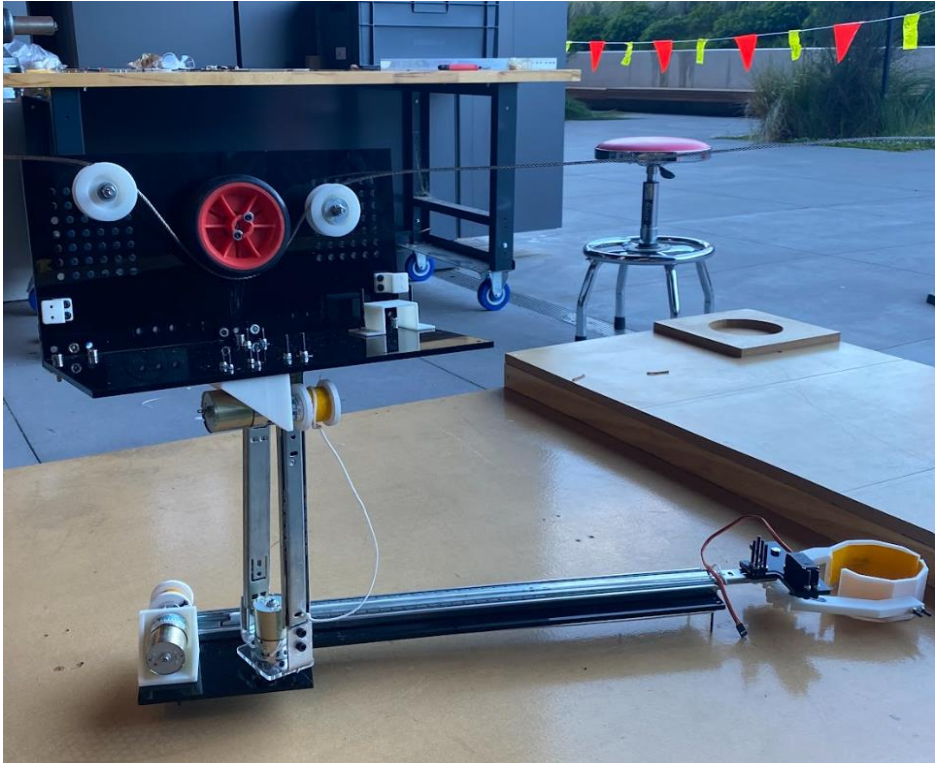


Figure 37: Assembling the systems together in accordance with our new CAD



Figure 38: Shifting the drawer slides through the plate to allow for the system to properly actuate and not touch the Chasm floor.

2.4 Design Comparison - Second Major Difference (Word Count: 374)

The second major difference between our Preliminary CAD Concept and our final manufactured device is that in our preliminary concept we used a rack and pinion mechanism to extend our claw but in our final manufactured device we used drawer slides and spool which used almost the same mechanism that was used for the vertical actuation mechanism. During the preliminary design we went for the rack and pinion system as it would have given us greater control on the amount of extension and retraction. One of the oversights of this mechanism is the friction. The rack sits flush on the base plate as shown in Figure 39, this results in a great amount of friction when the rack is moved and there is no rolling mechanism in place to reduce the friction. The team then started to look for alternative ideas that can replace the current mechanism or overcome the problem of friction. The team then decided to go with the spool and drawer slides mechanism as shown in Figure 40 and Figure 41. The drawer slides use ball bearings to move the slides which encounters less friction, this means we could get away with a motor with less torque than the one that would have been used in the rack and pinion mechanism. Later we realised our system only needs to extend twice and no retraction is necessary which means a string connecting the drawer slides and the spool to extend the slides should fulfil the objective. As we were using almost the same mechanism for the vertical actuation we used the same spool design for both of the mechanisms. During our testing phase we found out we had another oversight which was balance. Our system was not balanced when it was suspended on the wire fully. Placing the motor and spool on the back end of the drawer slides made that side too heavy and caused it to tilt. We then decided to move the motor and the spool to the other end of the drawer slides. Figure 40 shows the system when it was unbalanced and Figure 41 shows the new position of the motor and spool. This new configuration solved the counterweight problem and made it into the final design.

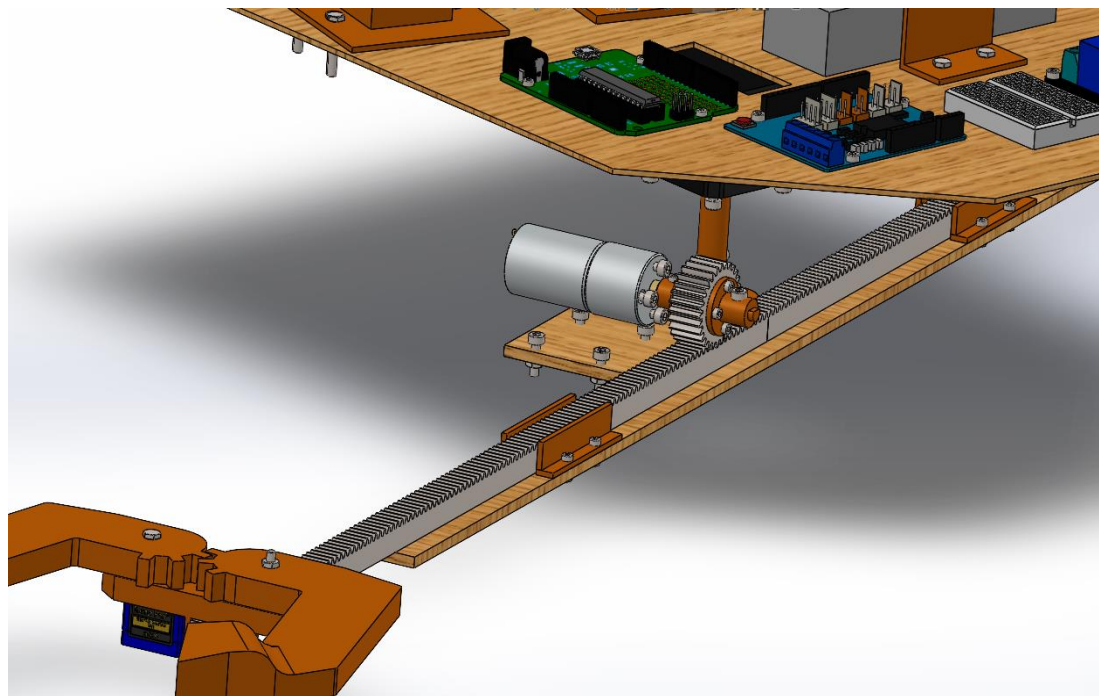


Figure 39: CAD of preliminary concept where rack and pinion is used and the rack sits directly on top of the base plate

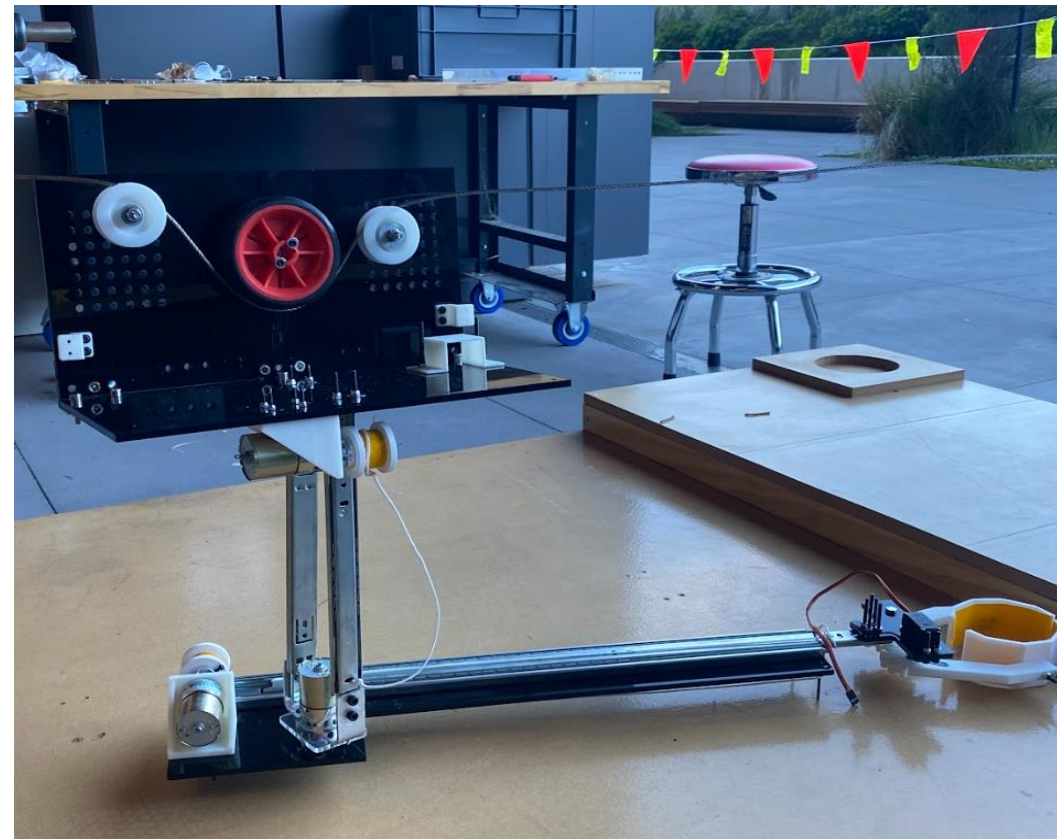


Figure 40: The position of the motor when the system was unbalanced.

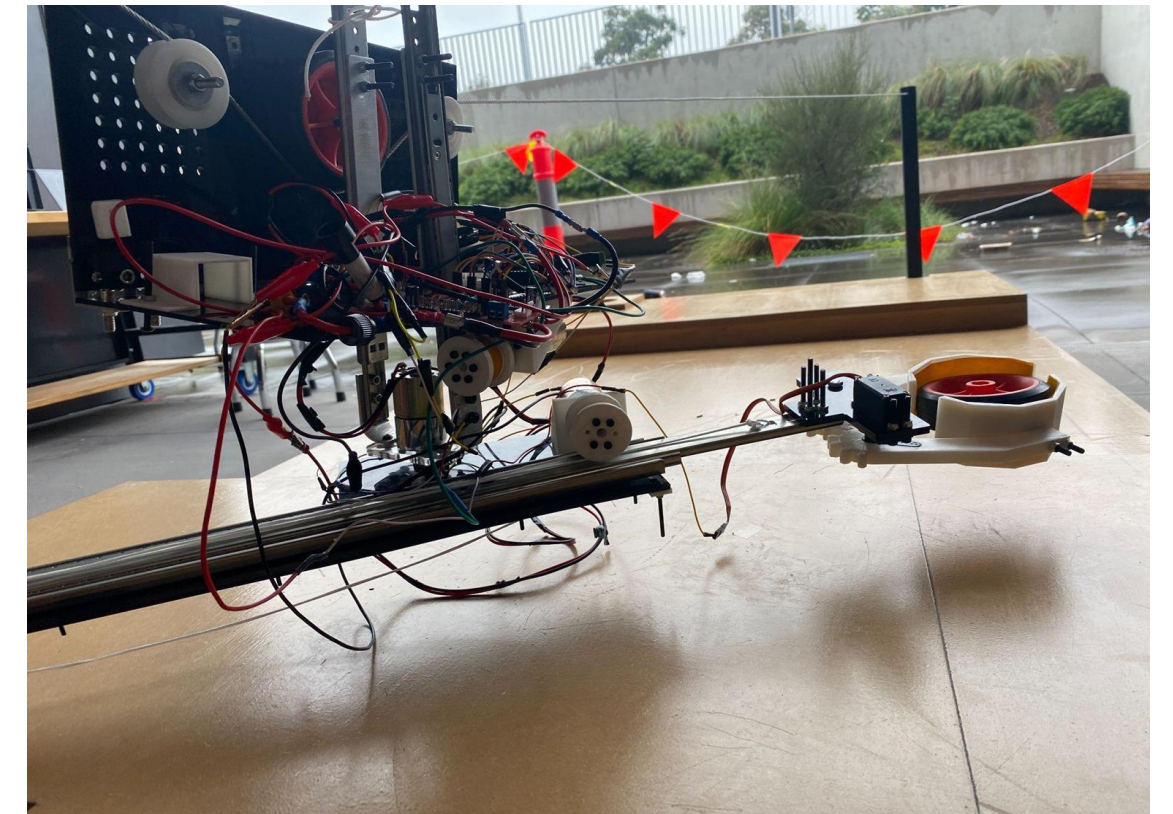


Figure 41: The motor and the spool shifted to the front end of the drawer slides.

3 Project Reflection

What went right or wrong, and what would you have done differently if you were given the chance?

What went right was that the team was able to design, develop and test the proposed prototype. The team also communicated well and planned the project as best as they could with limited experience in working with robotics.

The team ran into many issues before building the system. To begin with, the team designed their system with a linear actuator for the vertical system to raise and lower the system and rack and pinion for the horizontal system to extend and retract the arm to pick up the package and deliver to a certain zone. They thought that would be good for their system when the parts were integrated. However, the team decided to change those components after submitting their first report as they thought that the teeth of the gear would be difficult to construct to rotate easily and smoothly through the rack.

Furthermore, the team was not able to go with linear actuators because they were found to be costly. The team decided to go with a spool and drawer slide for both the horizontal and vertical systems as they would be easy to construct and work with when the parts were put together.

However, the team had many issues when integrating the subsystems. First, the team had some issues with the wire traversal as the position of the two pulleys was not horizontal. This affected the whole system to travel across the Chasm efficiently. Additionally, the team had a counter weight issue. Even with a higher stall torque of the purchased motor, the rotation system wasn't able to rotate the horizontal arm when the system was lifted. Moreover, the team had an issue with the claw mechanism as it didn't hold the package for quite a long time.

If the team had the chance to do the competition one more time, they would have fixed their counter weight issue early enough in order to ensure that the whole system would have run smoothly and effectively.

Did you start early enough? How effectively did you manage your team?

Even though, the team had most of the resources in the lab to start building their system, and they used Trello, Gantt Chart and Messenger to manage their tasks since each teammate knew what they had to do for their subsystem, the start date was not early enough as each teammate had other important tasks to complete for other units, and all of them had work. Moreover, the team was able to order their motors for each subsystem before the mid-semester break. However, the team got their motors on Tuesday in week 8.

Furthermore, the team took most of week 9 soldering their motors and boards (electrical components) to connect them to their system as a final phase of integration. Therefore, the team had only 3 days to start testing the whole system. However, it didn't go as the team thought it would, as they had some issues with their motors not running sequentially when the button was pressed. The team had to be in the testing area until 4 in the morning to fix that issue, but it couldn't be fixed at that time. The team would have fixed most of the problems of their system if they had started early to get a good mark for their Warman Competition.

3 Project Reflection

What lessons have you learned as a team that you might be able to apply to future project work? What have you learnt about the difference between theory and practice?

The team has gained some incredible experience throughout the project in areas including, but not limited to, mechanical and electrical design and engineering, project planning and the importance of consistently updating/reflecting on the timeline, teamwork/management, communication, leadership and testing/development.

All members of the team had experience working in group projects, at the university and for external student teams. This meant that the team members were able to set clear expectations for communication, teamwork and project planning/management from the start. As the project progressed the team realised the importance of maintaining these standards, as a momentary break for team members meant the project fallen behind, thus, they had a very tight schedule. Although they could devote extra time to the project to make up for this, it was to the detriment of their performance in the long run as they kept breaking their own standards, meaning they kept having to devote resources to fixing things that should have already been solved.

The team had also learnt that it would require more time to be devoted to designing/analysing/constructing/testing the areas of the build where there were knowledge gaps in order to successfully complete such a project. As the team found out the hard way, having a basic understanding the relevant theory in a certain area is much different than being able to apply the theory into practice. For example, as the project members had little knowledge interacting with electrical systems, more time should have been devoted to researching, designing, analysing and testing these electrical components prior to construction of the build.

What have you collectively learnt about working with and managing a team of people?

Overall, the team learnt that the key to working in a project with other team members falls down to honesty. This does not just fall under communication however, honesty about all aspects of everyone's skillset. If there is a clear lack of understanding in a certain area, the team needs to be able to identify that early, and devote more time to get adequate resources to ensure it does not become the downfall of the project.

4 Reconciled Project Expenses

The final project cost was: \$244.31.

To make sure everyone was on the same page with spending, we made a team rule that no one can purchase any materials without running it by through the team and trying to justify costs. As a result, when someone did buy any materials (after getting approval from the team), immediately after, a receipt, a breakdown of costs per team member, and account details of the purchaser were sent to the team group chat and remaining costs were transferred instantaneously. Hence, all team members have fully contributed to the final cost.

Additionally, the team took care of the Arduino Kit’s provided. As a result, the Arduino Kit’s were returned with no broken or missing items. Therefore, no costs due to neglect or recklessness of the Arduino Kit are required or accounted for.

Team #	21	Team Members	Jason	Ryan	Faisal	Shafin	\$244.31
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			Proportion paid by each Team Member				Notes
#	Item(s) Description	Price	1	2	3	4	
1	1 x Set of Drawer Slides 5000x46mm	\$16.17	0.25	0.25	0.25	0.25	Bought by Jason Abi Chebli from Bunnings Warehouse Notting Hill on 12/04/2022. Payment's have been recieved from all team members.
2	1 x Antislip Tape Brutus	\$7.50	0.25	0.25	0.25	0.25	Bought by Jason Abi Chebli from Bunnings Warehouse Notting Hill on 12/04/2022. Payment's have been recieved from all team members.
3	1 x Rope Utility Cord Grunt 3mmx20m	\$5.30	0.25	0.25	0.25	0.25	Bought by Jason Abi Chebli from Bunnings Warehouse Notting Hill on 12/04/2022. Payment's have been recieved from all team members.
4	1 x 95mm Black Rubber Type	\$5.10	0.25	0.25	0.25	0.25	Bought by Jason Abi Chebli from Bunnings Warehouse Notting Hill on 12/04/2022. Payment's have been recieved from all team members.
5	2 x 50mm Nylon Pully Tyres	\$10.00	0.25	0.25	0.25	0.25	Bought by Jason Abi Chebli from Bunnings Warehouse Notting Hill on 12/04/2022. Payment's have been recieved from all team members.
6	1 x Set of Drawer Slides 250x25mm	\$7.53	0.25	0.25	0.25	0.25	Bought by Jason Abi Chebli from Bunnings Warehouse Notting Hill on 12/04/2022. Payment's have been recieved from all team members.
7	1 x Nylon Fishing Line Rope	\$8.00	0.25	0.25	0.25	0.25	Bought by Jason Abi Chebli from Bunnings Warehouse Notting Hill on 12/04/2022. Payment's have been recieved from all team members.
8	1 x Metal DC Geared Motor w/Encoder - 12V 251RPM 18Kg. cm	\$45.65	0.25	0.25	0.25	0.25	Bought by Ryan Hartshorne from Core Electronics (online) on 13/04/2022. Payment's have been recieved from all team members.
9	2 x Pololu Universal Aluminum Mounting Hub for 6mm Shaft, M3 Holes (2-Pack)	\$25.00	0.25	0.25	0.25	0.25	Bought by Ryan Hartshorne from Core Electronics (online) on 13/04/2022. Payment's have been recieved from all team members.
10	3 x Metal DC Geared Motor - 12V 100RPM 42kg.cm	\$64.20	0.25	0.25	0.25	0.25	Bought by Ryan Hartshorne from Core Electronics (online) on 13/04/2022. Payment's have been recieved from all team members.
11	5 x Snap-Action Switch with 16.7 mm Lever: 3-Pin, SPDT, 5A	\$9.75	0.25	0.25	0.25	0.25	Bought by Ryan Hartshorne from Core Electronics (online) on 13/04/2022. Payment's have been recieved from all team members.
12	Express Shipping Cost for CoreElectronic's Parts	\$23.06	0.25	0.25	0.25	0.25	Bought by Ryan Hartshorne from Core Electronics (online) on 13/04/2022. Payment's have been recieved from all team members.
13	3 x 7806 Voltage Divider 6V 1A	\$5.40	0.25	0.25	0.25	0.25	Bought by Jason Abi Chebli from Altronics Springvale on 04/05/2022. Payment's have been received from all team members.
14	1 x 10 Pcs Crocodile Clip Set	\$11.65	0.25	0.25	0.25	0.25	Bought by Jason Abi Chebli from Altronics Springvale on 04/05/2022. Payment's have been received from all team members.
15							

Calculations			
1	2	3	4
4.04	4.04	4.04	4.04
1.88	1.88	1.88	1.88
1.33	1.33	1.33	1.33
1.28	1.28	1.28	1.28
2.50	2.50	2.50	2.50
1.88	1.88	1.88	1.88
2.00	2.00	2.00	2.00
11.41	11.41	11.41	11.41
6.25	6.25	6.25	6.25
16.05	16.05	16.05	16.05
2.44	2.44	2.44	2.44
5.77	5.77	5.77	5.77
1.35	1.35	1.35	1.35
2.91	2.91	2.91	2.91
0.00	0.00	0.00	0.00

\$61.08	\$61.08	\$61.08	\$61.08
---------	---------	---------	---------

#	Team Member	Contributed	Total Owning (-), Owed (+)	Action?
1	Jason	\$61.08	\$0.00	No Action Required
2	Ryan	\$61.08	\$0.00	No Action Required
3	Shafin	\$61.08	\$0.00	No Action Required
4	Faisal	\$61.08	\$0.00	No Action Required
	Average Contribution	\$61.08		

5 Conclusions

Although the system was not able to successfully deliver the package to Deposit Zone D as initially designed, the team learnt a tremendous amount throughout the project. Working with every member on the team has been really fun and it has been a great experience to work alongside like minded people to complete a common goal. Although this project took a lot out of us, we really appreciated the opportunity to complete such a task. An honest recommendation would be for this unit to be worth double credit points. We say this not because we put double the amount of time and effort we had for any other unit just to get some result in the Warman Competition, but because we all believed that if this extra time was not devoted, we would not have learnt nearly a quarter of what we were able to. We feel that the project gave us so much opportunity and exposure to the world of robotics and we really appreciate it.

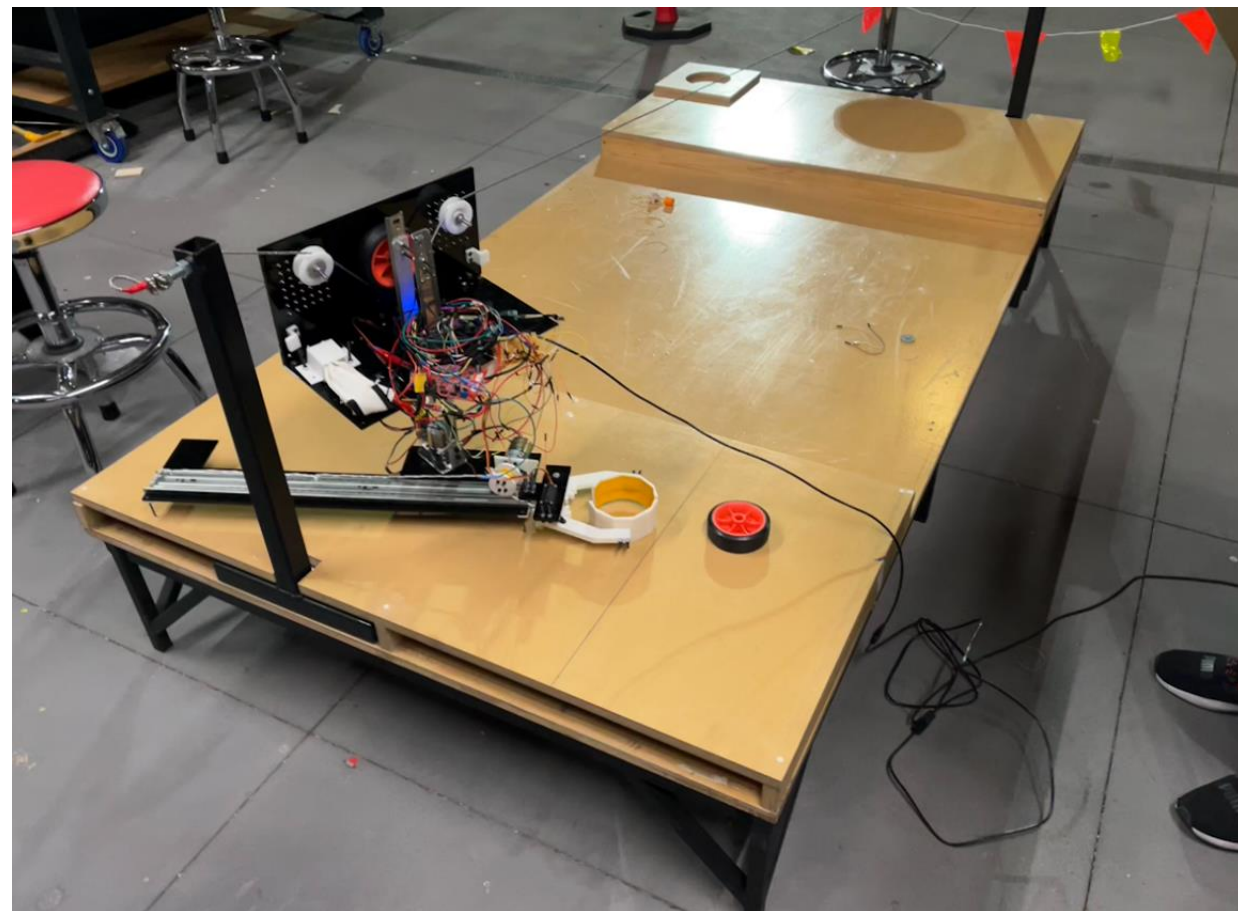


Figure 43: Testing our system on the track.

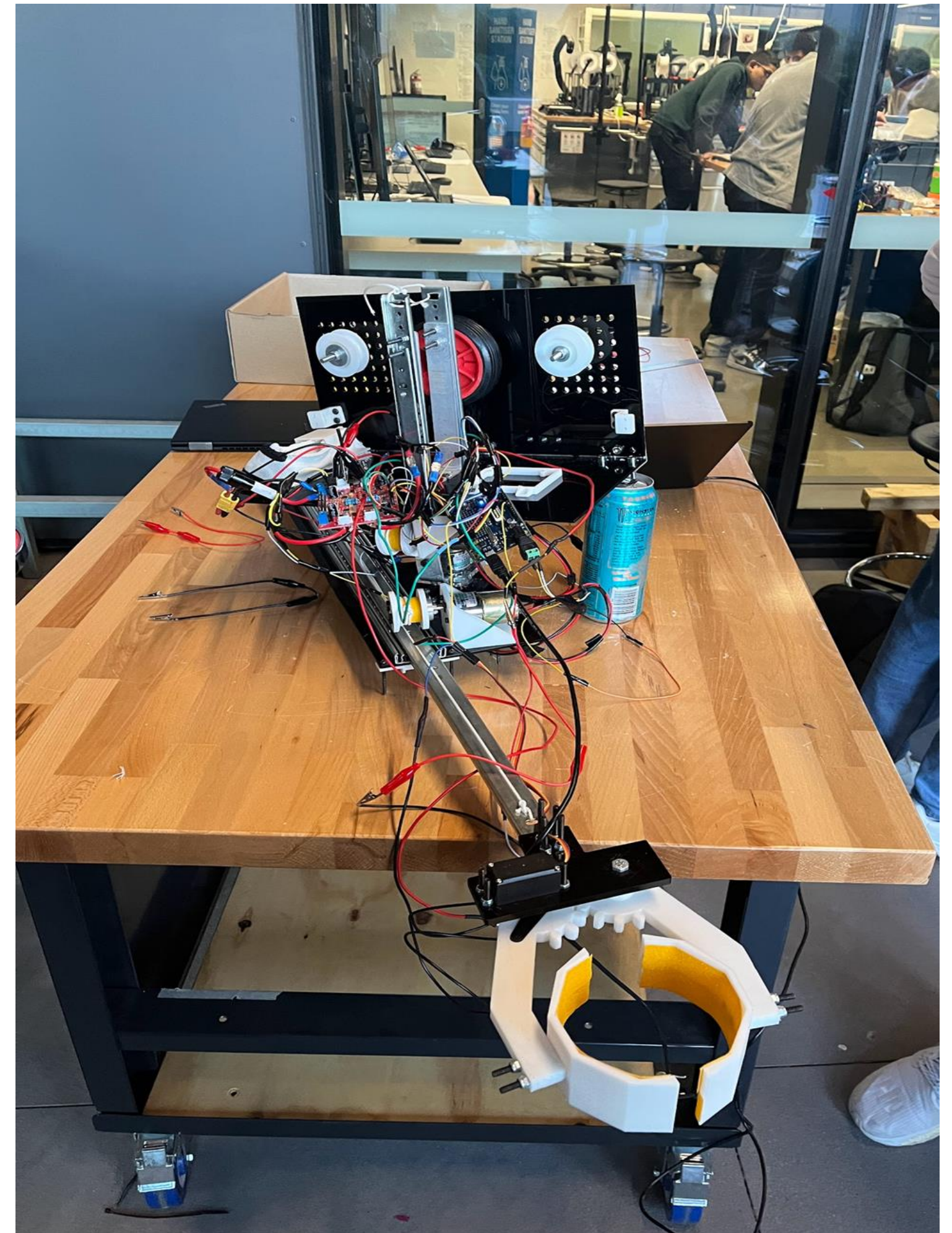
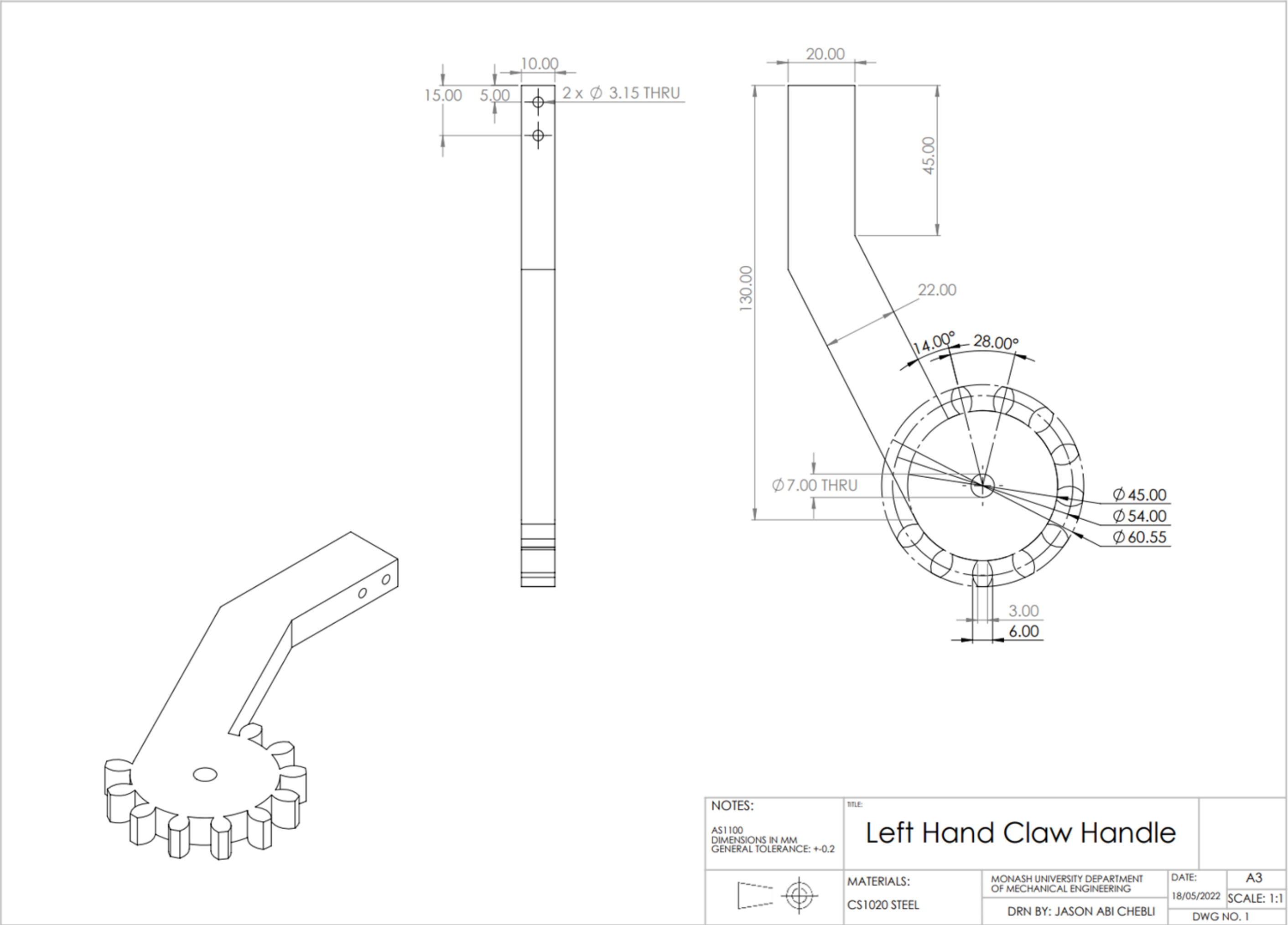
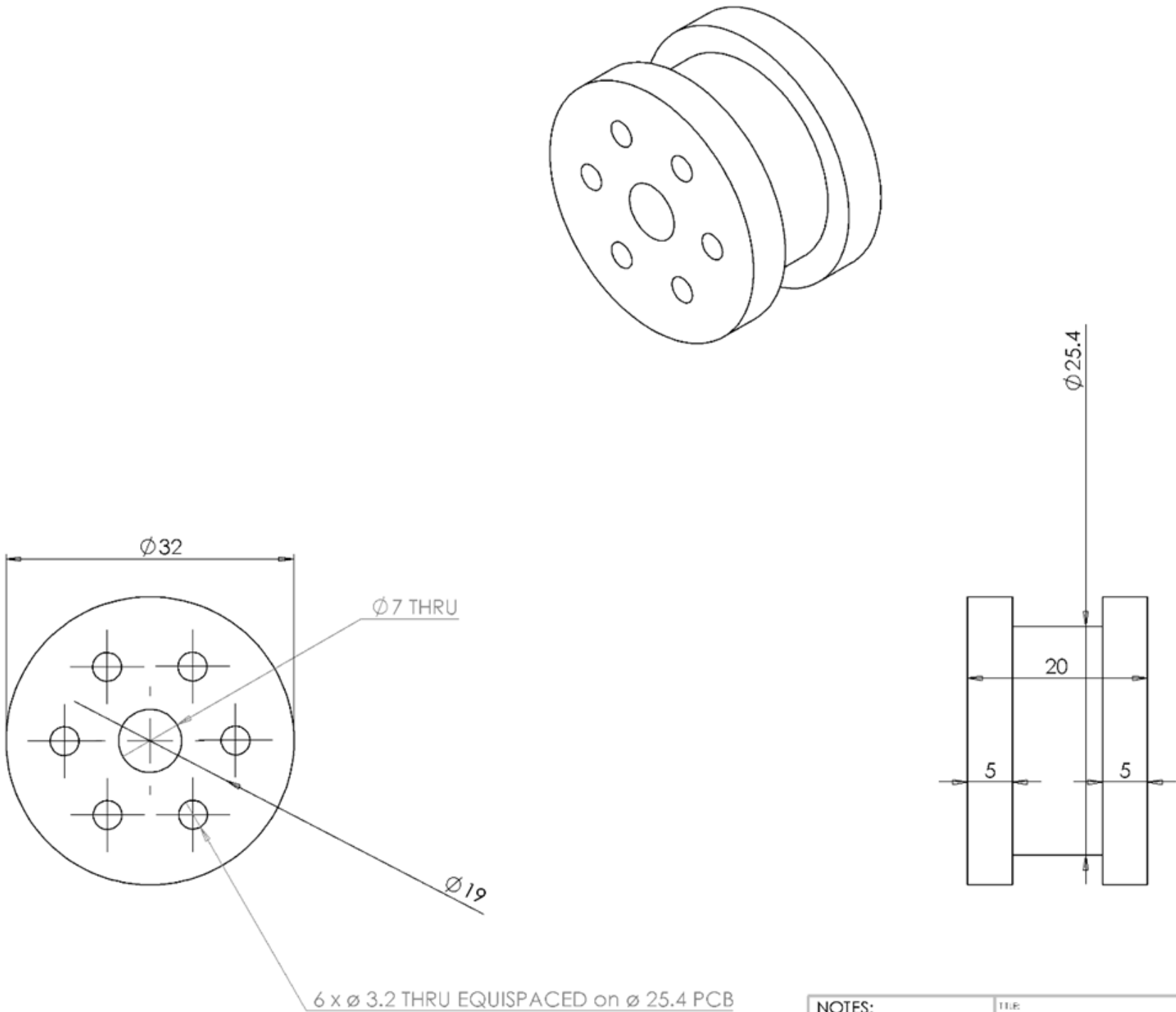


Figure 42: The manufactured system after competition.

Appendix A: Detailed Drawing

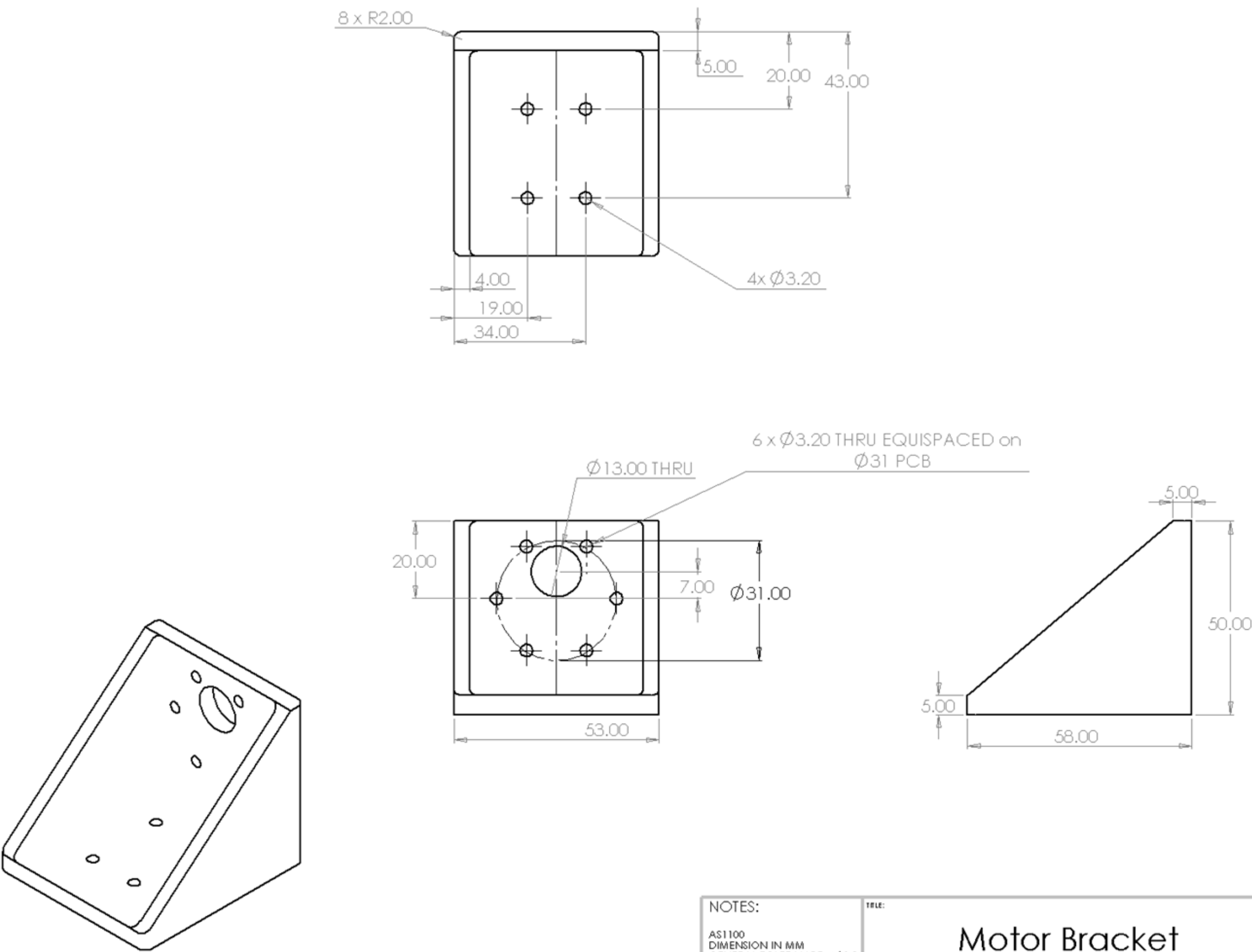


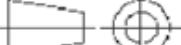
Appendix A: Detailed Drawing



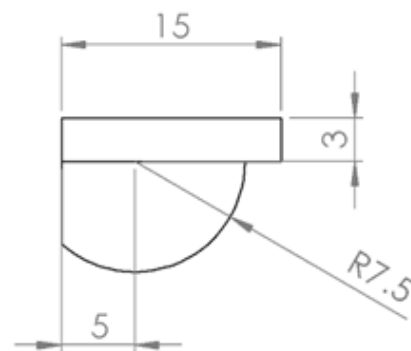
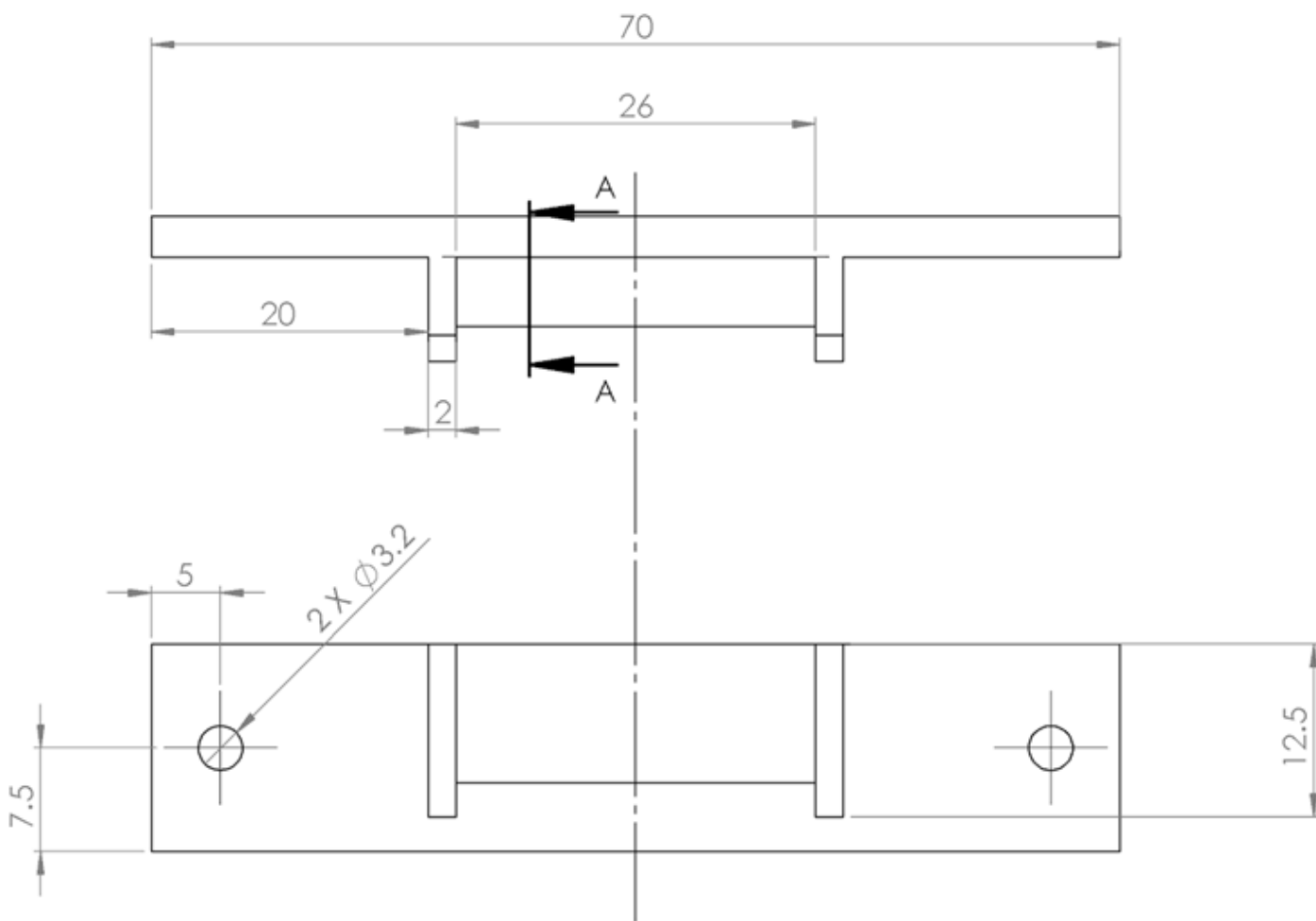
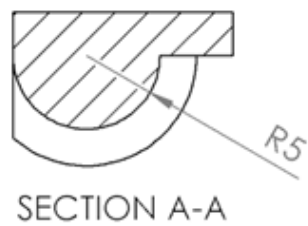
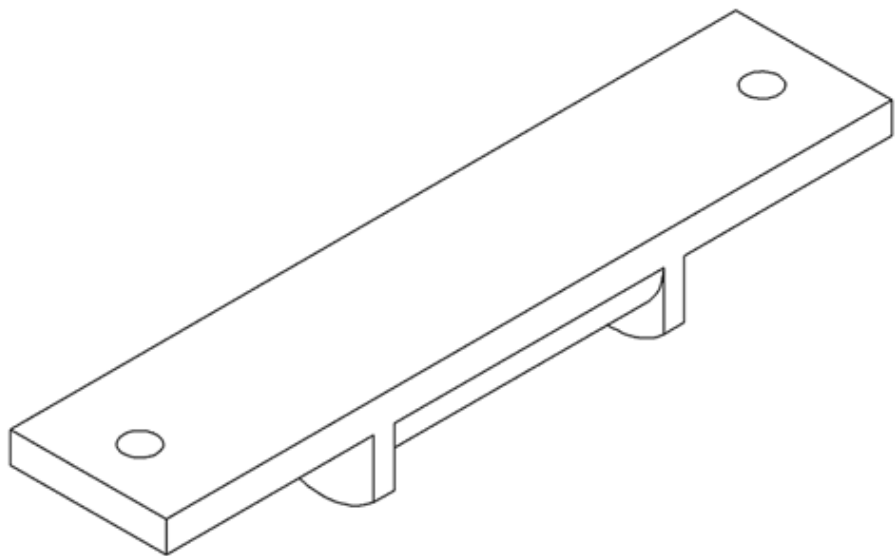
NOTES: AS1100 DIMENSIONS IN MM GENERAL TOLERANCE: ± 0.2		TITLE: DC Motor Spool			
		MATERIALS: CS1020 STEEL		DATE: 23/03/2022	
		MONASH UNIVERSITY DEPARTMENT OF MECHANICAL ENGINEERING		A3	
		DRN BY: RYAN HARTSHORNE		SCALE 2:1	
				DWG NO. 2	

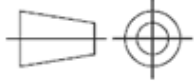
Appendix A: Detailed Drawing



NOTES: AS1100 DIMENSION IN MM GENERAL TOLERANCE: +/-0.2		TITLE: Motor Bracket			
	MATERIALS: CS1020 STEEL		MONASH UNIVERSITY DEPARTMENT OF MECHANICAL ENGINEERING		DATE: 19/05/22
			DRN BY: Faisal Almaslukh		A3
					SCALE: 1:1
				DWG NO. 3	

Appendix A: Detailed Drawing



NOTES: AS1100 DIMENSIONS IN MM GENERAL TOLERANCE: +/- 0.2	TITLE: Wire Guide			wire guide	
	MATERIALS:	MONASH UNIVERSITY DEPARTMENT OF MECHANICAL ENGINEERING	DATE: 23/05/2022	A3	
	CS1020 STEEL	DRN BY: MAHDIR SHAFIN HASAN ID: 32236093	SCALE: 2:1		
	DWG NO. 4				

Appendix B: Team Meeting Minutes - Week 3

MEC2402 / MAE2412		TEAM MEETING 01					DATE	15/03/2022	START	2:35 PM
TEAM #	21	ICE BREAKER, CONTACT DETAILS & COMMUNICATIONS					VENUE	LG03	FINISH	3:45 PM
1: ATTENDANCE		3: TEAM CONTACT DETAILS (5 mins)					4: ICEBREAKER INFORMATION (10 mins)			
ON-TIME		TEAM MEMBER NAME (& Nickname?)		STUDENT EMAIL AND PHONE NUMBER			Introduce the team mate next to you. SCRIBE (only) to make some brief notes			
Faisal Ibrahim M Almaslukh		Faisal (Felly)		404524829			Faisal, or Felly, from Saudi Arabia originally, did high school in melbourne, picked up his accent from a friend. good at coding. Interested in growing and roasting coffee beans.			
falm0005@student.monash.edu										
Ryan Hartshrone		Ryan		459141091			Ryan, born in Camberwell, Greek from mother's side, English from father's side. Grandmother likes tea, from MDN.			
Jason Abi CHebli				rhar0031@student.monash.edu						
Mahdir Shafin Hasan		Shafin		406551067			Shavin, grew up in Bangladesh, he has a roommate that likes to party so he joins in with him. Lots of partying. Good knowledge of coding (python, js, matlab, java), fast-learner. Activities department of MUISS.			
				mhas0039@student.monash.edu						
		Jason		439345328			Jason's family is from Lebanon. He moved to South Africa when he was very young, his first language is German and moved to Australia when he was 7. He is a part of Nova Rover, studying Mechanical Engineering.			
APOLOGIES				jabi0003@student.monash.edu						
Nathan Mudappathi Nair		Nathan		468582444						
				nmud0004@student.monash.edu						
LATE										
		5: COMMUNICATION & FILE SHARING (10 mins)					6: TEAM WORKING TIMES (5 mins)			
		COMMUNICATIONS: How will your team talk informally on a day-to-day basis for updates and logi					When can the team ALL work together or review progress on the projects?			
NO SHOWS		Discuss the options:	Emails?	Facebook?	SMS?	Phone Calls?	We suggest trying to find an additional 2-4 hours free time on either Monday or Tuesday outside of your tute or lab time, so that you can work together in Woodside. You can also indicate times members are busy or working etc. Try to decide where you will meet to work.			
		Your Primary Choice and Why?:	Messenger Group Chat for Instant DM, easily accessible and messaging is instant for project work. We also use it daily							
		Your Back-Up option?:	Email				Selected Time(s)	Thursday 4:30-6:30; Second Option (if we require more time) : Wednesday 10-12		
		FILE SHARING: How will your team share project files?						AM Options		PM Options
2: MEETING ROLES		Discuss the options:	Drive?	Facebook?	Emails?	Other?	Mondays			
TIMEKEEPER		Your Primary Choice and Why?:	Google Drive to ensure a central repository to store all information.				Tuesdays			
Shafin		Your Back-Up option?:	Email, as you have a receipt of everything sent				Wednesday	10:00-12:00	3:00-6:00	
SCRIBE		URGENT CONTACT: If the need is urgent, how should team members contact each other?					Thursdays	4:30 - 6:30		
Ryan		Discuss the options:	Emails?	Facebook?	SMS?	Phone Call?	Fridays			
FACILITATOR		Your Primary Choice and Why?:	SMS because people will generally check sms more than messenger				Saturdays	5:00-9:00		
Jason		Your Back-Up option?:	Phone call as last resort in urgent situation				Sundays	3:00-6:00		

Appendix B: Team Meeting Minutes - Week 4

MEC2402/MAE2412		WEEKLY STAND-UP MEETING		DATE	Week 4	START	2:45 PM	TEAM #	21
Team Member	What you planned to do in the last week? (cut and paste VERBATIM from last week's minutes)	What did you achieve last week?	What you plan to do in the this thursday?	Any impediments preventing you from achieving these goals?					
Faisal	Come up with creative solutions for the system . Choose and do a sketch of a design of your idea/solution.	Came up with some alternative solutions to the system and thought of and sketched a few different system ideas.	Introduction, Factors, Isometric Concepts Sketches	Another unit's assignment due next wednesday					
Shafin	Come up with creative solutions for the system . Choose and do a sketch of a design of your idea/solution.	Brainstormed some system ideas and did a sketch of a system idea.	Function Chart, Requirements and Specifications Table, Isometric Sketches	Two submissions on friday					
Jason	Come up with creative solutions for the system . Choose and do a sketch of a design of your idea/solution.	Came up with creative solutions for the system . Choose and do a sketch of a design of your idea/solution. I also started CADings. I also looked into and research arm.	Creative solution generation grouping, Creative solution generation table, isometric sketches, Decision making statistical significance	No					
Nathan	Come up with creative solutions for the system . Choose and do a sketch of a design of your idea/solution.	Thought about solutions.	Factors, Effects, Isometric Concept Sketches	Another unit's assignment due next wednesday					
Ryan	Come up with creative solutions for the system . Choose and do a sketch of a design of your idea/solution.	Came up with creative solution for the system.	Objectives, Isometric Concept Sketches, Decision Making Table/Evaluation, Conclusion	Another unit's assignment due next monday					

Appendix B: Team Meeting Minutes - Week 5

MEC2402 / MAE2412		TEAM MEETING		DATE	29/03/2022	START TIME	2:30 PM
TEAM #	21	Team Work Reflection		VENUE	LG03	FINISH TIME	3:30 PM
1: ATTENDANCE		4: DISCUSSION TOPIC	Rate your team, on work to date	What was done WELL by the TEAM?	WHAT needs improvement?	HOW will the team IMPROVE?	
ON-TIME		QUANTITY of Work Completed (by the whole team)	9/10	Were able to finish all the required slides.	We met the requirements, however we feel as though we could have completed additional drawings that would have assisted the report.	In future, we will ensure there is enough time to allow us to exceed the quantity requirements and expectations.	
Ryan							
Jason							
Shafin							
Faisal							
		QUALITY of Work Completed (by the whole team)	8/10	Everyone completed their work to a high standard prior to review from the team, ensuring there was very few revisions required.	Content backed up with more research to back up findings. More feedback required from the whole team rather than certain individuals who felt more confident with certain aspects of the report.	Improving confidence and skills in all areas. Implementing a dedicated feedback section where work will be finalised prior to the meeting and then only under review in the feedback section. Could also implement a buddy system where two people review each others work.	
APOLOGIES		COMMUNICATION & INTERACTION between team members	6/10	We were able to attend meetings and utilise the google drive and messenger in order to communicate and illustrate our ideas.	We need to ensure each week every member is committed, able and willing to continue to contribute to the project. Need to check in with everyone to see how members are going completing tasks, and team members need to communicate earlier if they are struggling with a task so that they can receive appropriate help. Also making sure people update the google drive contents regularly.	Bi-weekly messenger checkups to see how everyone is faring with their allocated tasks.	
LATE							
		SETTING & MEETING deadlines	8/10	We utilised meetings outside of class and kept an online Trello Board and Gantt Chart to manage the tasks required for the project at each stage. Everyone completed their tasks on time because of this.	More aggressive deadlines and/or setting deadlines earlier so we know in advance what is required in more detail.	Breaking down the task at hand earlier.	
NO SHOWS							
		Level of KNOWLEDGE & SKILLS (of the whole team)	7/10	As a team we leveraged our strengths and ensured the people who had the best skills equipped to tackle certain tasks were completing these tasks.	The whole teams competency in their weaknesses needs to be improved so that they can aid others in areas they are weaker in if we ran into time constraints.	Try to upskill teammembers in areas they are weaker in.	
2: MEETING ROLES							
TIMEKEEPER							
Shafin		RESULTS & OUTCOMES (of the whole team)	10/10	Although we have an average of about 7.6/10, we believe our standards are incredibly harsh and quite high, and our report was of incredible detail and quality.	We definitely have areas of improvements, we need to upskill our members in areas they are weakest, enhance communication about tasks and work on ensuring we have a great team culture that is open to sharing.	Ensure team members are completing weekly CAD tasks and other aspects of the unit.	
SCRIBE							
Ryan							
FACILITATOR							
Jason							

Appendix B: Team Meeting Minutes - Week 6

MEC2402/MAE2412	
TEAM #	21
1: ATTENDANCE	
ON-TIME	
Jason	
Shafin	
Faisal	
Ryan	
APOLOGIES	
LATE	
NO SHOWS	
2: MEETING ROLES (2 mins)	
TIMEKEEPER	
Shafin	
SCRIBE	
Faisal	
FACILITATOR	
Jason	

ITP Metrics PEER EVALUATION REFLECTION					
TEAM NAME		Team 21			
3: COMPLETION? (5 mins)					
TEAM MEMBER NAME	Survey done?	Feedback Reviewed?	Factor Reviewed?	Satisfaction? 1 - 10	
Jason	Yes	Yes	Yes	9	
Shafin	Yes	Yes	Yes	9	
Faisal	Yes	Yes	Yes	8	
Ryan	Yes	Yes	Yes	9	
3: TEAMWORK IN THIS GROUP COULD BE IMPROVED BY		Pairing people together and splititng up tasks amongst pairs.			
		Communication to the team via messenger.			
		Stricter dedication to trello and making sure to continuously update it.			

DATE	05/04/2022	START	2:20
VENUE	LG03	FINISH	2:45
4: POSITIVE CONTRIBUTION (10 mins)			
TEAM to nominate ONE THING each member did well			
Jason was great at the CAD related work and even better as a cheerleader.			
Shafin was good at asking questions and contributed well to the project.			
Faisal (Felly) was initiative at taking the big tasks, and he was opened to accept feedback from the teammates			
Ryan was opened to feedback to improve his drawings.			
6: MOST COMMON AREA FOR IMPROVEMENT (2 mins)			
Communication			
HOW TO IMPROVE:	By reviewing the work before the meetings and preparing for them as well.		

5: AREA FOR IMPROVEMENT (10 mins)	
INDIVIDUALS to nominate ONE THING they can improve	
He could've shared the workload with the team.	
He should've taken inititive for the big tasks.	
He should've communicated more with the team	
He should've been more clear and direct with his communication. Also, he should've come early to the team meetings.	

Appendix B: Team Meeting Minutes - Week 7

MEC2402	WEEKLY TEAM MEETING MINUTES (and review by teaching staff)	Meeting Date	12/04	Meeting Time	2:30
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Team members:	Jason, Ryan, Faisal, Shafin
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Review of the Last Week					
Copy and Paste from last week's minutes				Team to Meet and Update BEFORE Workshop	
# or Set	Description of Task	Measurable Deliverable(s)	Team Members (Lead / Supporting)	Notes on review of Deliverables	Delivered? (%)
1.1	Create a budget that includes the materials for the horizontal, vertical arm and rotation system	Present budget to the team and have an excel sheet available	Faisal and Shafin	Need to work on the rotation system.	90%
1.2	Come up with initial design for the mechanism for the vertical and horizontal arm	Present design idea to team using Jamboard and any other tools necessary.	Faisal and Shafin	Need to finalise a design.	90%
2.1	Create a budget that includes the materials for the wire traversal, clamp and claw mechanism.	Present budget to the team and have an excel sheet available	Jason and Ryan	Done! Good job!	100%
2.2	Come up with initial design for the mechanism for the wire traversal, clamp and claw mechanism	Present design idea to team using Jamboard and any other tools necessary.	Jason and Ryan	Design is thought about, need to do some CAD to solidify ideas.	90%

Week	7
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Team	21
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Plan for the Next Week (ONLY)			
Team to Meet and Update BEFORE Workshop			
# or Set	Description of Task	Measurable / Tangible Deliverable(s) (what evidence is required to ensure the task has been completed?)	Team Members (Lead / Supporting)
1.1	Prototyping the vertical actuation system	Video and photo documentary in the group chat showing it.	Faisal and Shafin
1.2	Prototyping the horizontal actuation system	Video and photo documentary in the group chat showing it.	Shafin and Faisal
1.3	Prototyping the rotation system.	Video and photo documentary in the group chat showing it.	Shafin and Faisal
2.1	Prototype the wire traversal system.	Video and photo documentary in the group chat showing it.	Ryan and Jason
2.2	Prototype the clamp mechanism,.	Video and photo documentary in the group chat showing it.	Jason and Ryan
2.3	Prototype the claw mechanism,.	Video and photo documentary in the group chat showing it.	Jason and Ryan

Appendix B: Team Meeting Minutes - Week 8

MEC2402	WEEKLY TEAM MEETING MINUTES (and review by teaching staff)	Meeting Date	26/4/2022	Meeting Time	2:30
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Team members:	Jason, Ryan, Faisal, Shafin
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Review of the Last Week					
Copy and Paste from last week's minutes				Team to Meet and Update BEFORE Workshop	
# or Set	Description of Task	Measurable Deliverable(s)	Team Members (Lead / Supporting)	Notes on review of Deliverables	Delivered? (%)
1.1	Prototyping the vertical actuation system	Video and photo documentary in the group chat showing it.	Faisal and Shafin	Everythings been printed and cut. Just need to assemble and make sure it's appropriate.	80%
1.2	Prototyping the horizontal actuation system	Video and photo documentary in the group chat showing it.	Shafin and Faisal	Everythings been printed and cut. Just need to assemble and make sure it's appropriate.	80%
1.3	Prototyping the rotation system.	Video and photo documentary in the group chat showing it.	Shafin and Faisal	Everythings been printed and cut. Just need to assemble and make sure it's appropriate.	80%
2.1	Prototype the wire traversal system.	Video and photo documentary in the group chat showing it.	Ryan and Jason	Everythings been printed and cut. Just need to assemble and make sure it's appropriate.	80%
2.2	Prototype the clamp mechanism,.	Video and photo documentary in the group chat showing it.	Jason and Ryan	From prototyping we decided that this feature is no longer require this task/feature. Hence, it was scrapped.	N/A
2.3	Prototype the claw mechanism,.	Video and photo documentary in the group chat showing it.	Jason and Ryan	Everything has been complete!!	100%

Week	8
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Team	21
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Plan for the Next Week (ONLY)			
Team to Meet and Update BEFORE Workshop			
# or Set	Description of Task	Measurable / Tangible Deliverable(s) (what evidence is required to ensure the task has been completed?)	Team Members (Lead / Supporting)
1.1	Finalise vertical actuation system and combine it with the horizontal actuation system.	Video and photo documentary in the group chat showing it.	Faisal / Shafin
1.2	Mount motors and mounts to vertical and horizontal actuation systems and test motors work	Video and photo documentary in the group chat showing it. Motors should be able to rotate clockwise and anticlockwise.	Faisal / Shafin
2.1	Finalise claw system and combine it with the horizontal actuation system.	Video and photo documentary in the group chat showing it.	Jason / Shafin
2.2	Mount motors and mounts to claw systems and test motors work	Video and photo documentary in the group chat showing it. Motors should be able to rotate clockwise and anticlockwise.	Jason / Shafin
3.1	Finalise wire traversal system and combine it with the vertical actuation system.	Video and photo documentary in the group chat showing it.	Ryan / Shafin
3.2	Mount motors and mounts to wiral traversal systems and test motors work	Video and photo documentary in the group chat showing it. Motors should be able to rotate clockwise and anticlockwise.	Ryan / Shafin

Appendix B: Team Meeting Minutes - Week 9

MEC2402	WEEKLY TEAM MEETING MINUTES (and review by teaching staff)	Meeting Date	03/05	Meeting Time	2:30
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Team members:	Jason, Ryan, Faisal, Shafin
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Review of the Last Week					
Copy and Paste from last week's minutes				Team to Meet and Update BEFORE Workshop	
# or Set	Description of Task	Measurable Deliverable(s)	Team Members (Lead / Supporting)	Notes on review of Deliverables	Delivered? (%)
1.1	Finalise vertical actuation system and combine it with the horizontal actuation system.	Video and photo documentary in the group chat showing it.	Faisal / Shafin	Need to make L-bracket more L-shaped.	95%
1.2	Mount motors and mounts to vertical and horizontal actuation systems and test motors work	Video and photo documentary in the group chat showing it. Motors should be able to rotate clockwise and anticlockwise.	Faisal / Shafin	Ran out of time to finish motor testing. Need to make sure to do this.	45%
2.1	Finalise claw system and combine it with the horizontal actuation system.	Video and photo documentary in the group chat showing it.	Jason / Shafin	Done ! Look's fire!	100%
2.2	Mount motors and mounts to claw systems and test motors work	Video and photo documentary in the group chat showing it. Motors should be able to rotate clockwise and anticlockwise.	Jason / Shafin	Ran out of time to finish motor testing. Need to make sure to do this.	60%
3.1	Finalise wire traversal system and combine it with the vertical actuation system.	Video and photo documentary in the group chat showing it.	Ryan / Shafin	Pretty much done, just need to redo the L-bracket.	95%
3.2	Mount motors and mounts to wiral traversal systems and test motors work	Video and photo documentary in the group chat showing it. Motors should be able to rotate clockwise and anticlockwise.	Ryan / Shafin	Ran out of time to finish motor testing. Need to make sure to do this.	45%

Week	9
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Team	21
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Plan for the Next Week (ONLY)			
Team to Meet and Update BEFORE Workshop			
# or Set	Description of Task	Measurable / Tangible Deliverable(s) (what evidence is required to ensure the task has been completed?)	Team Members (Lead / Supporting)
1	Finalise remaining mechanical issues - counterweight issue.	Video and photo documentary in the group chat showing it.	Jason / Faisal
2	Finalise circuit diagram and make circuit boards.	Video and photo documentary in the group chat showing it.	Shafin / Ryan
3	Make wires, crimp and solder onto motors etc.	Video and photo documentary in the group chat showing it.	Jason /Ryan
4	Finalise code for individual and overall system.	Video and photo documentary in the group chat showing it.	Ryan / Shafin
5	Perform subassembly testing	Video and photo documentary in the group chat showing it.	Everyone
6	Perform whole-system testing	Video and photo documentary in the group chat showing it.	Everyone

Appendix B: Team Meeting Minutes - Note

In Addition to using the Meeting Minutes, the team also heavily utilised Trello and TeamGantt.

Trello allowed us to break down the work to the bare bones, assign people to tasks and track their progress. Everyone regularly updated Trello. One of our trello boards can be seen in Figure 44.

TeamGantt was also used to create a Gantt chart was was regularly updated every meeting to track the progress of the team and see if we were meeting deadlines. Our final Gantt chart can be seen in Figure 45.

In addition, regular updates were also provided on our team messaging group chat.

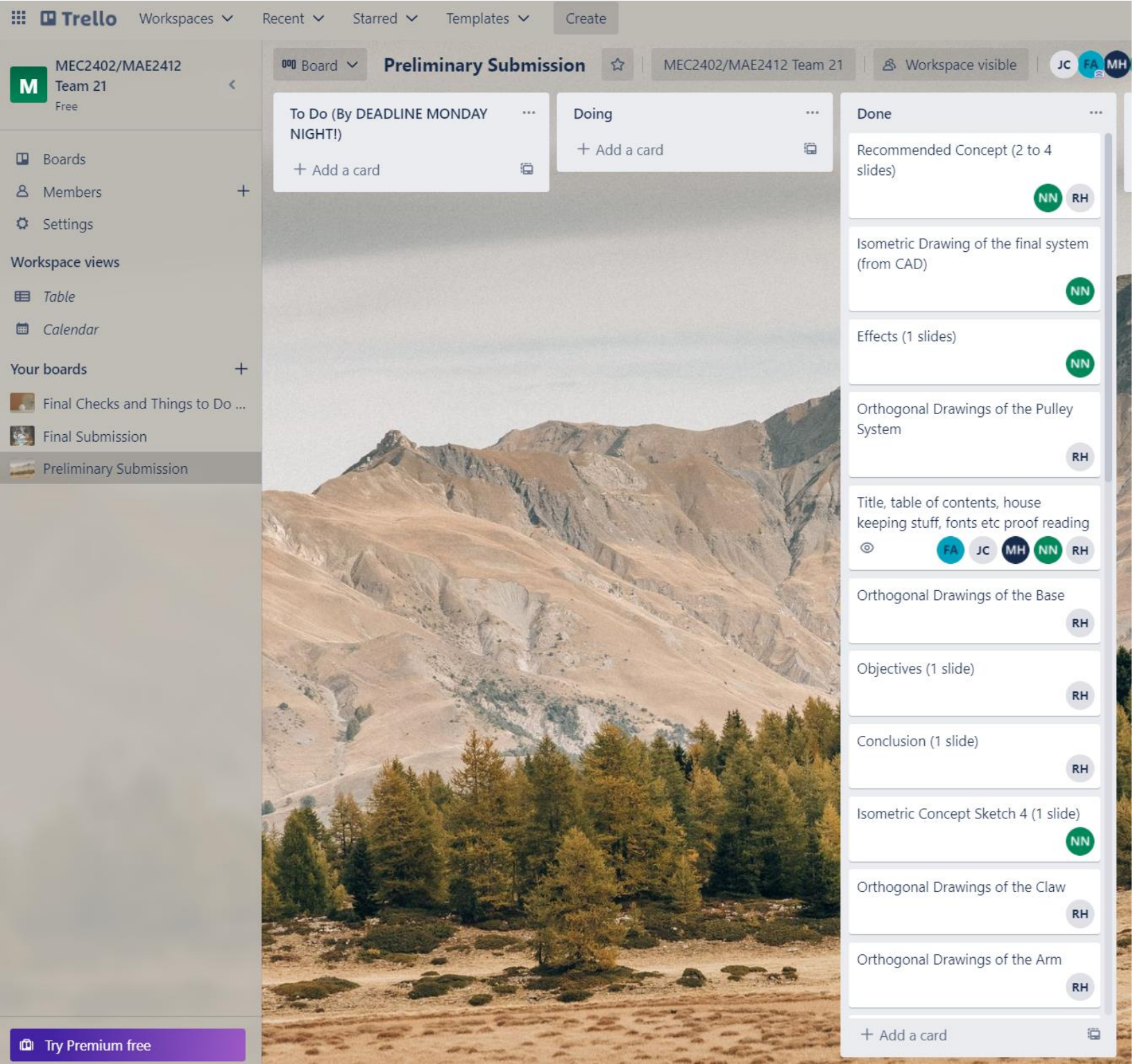


Figure 44: One of Team 21’s Trello Boards used to break down tasks further, assign work to individuals, set deadlines and track people’s progress.

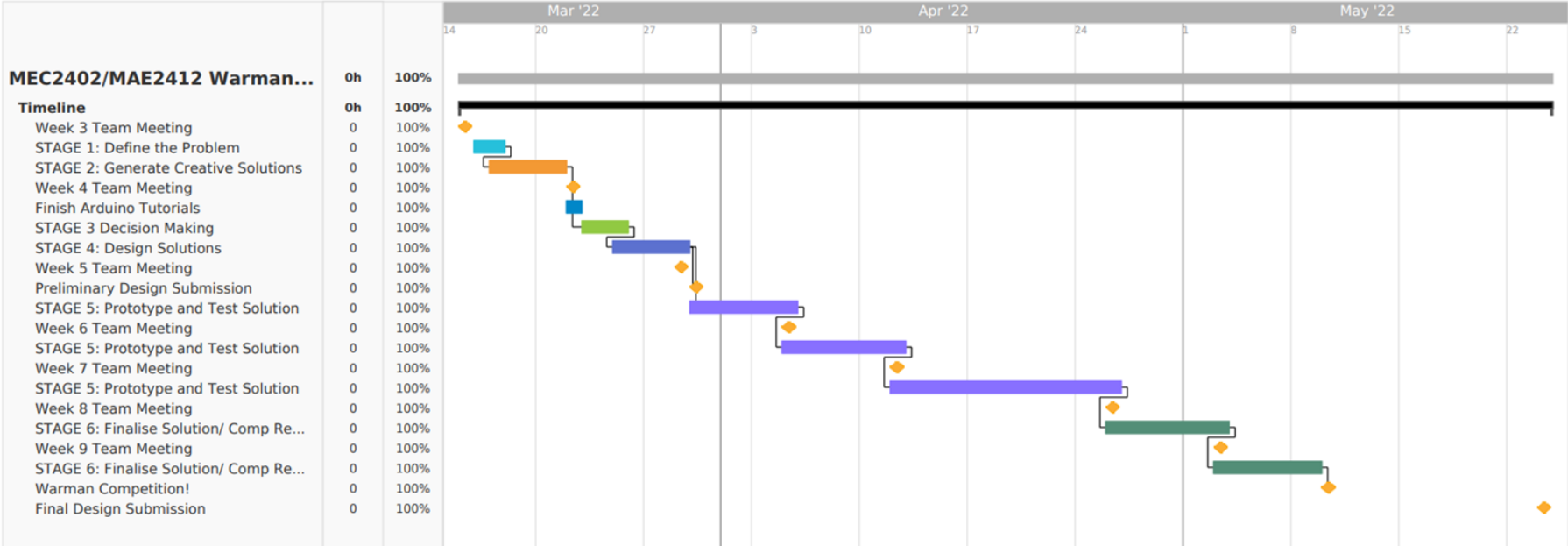


Figure 45: Team 21’s Gantt Chart that was regularly updated to track progress and deadlines.